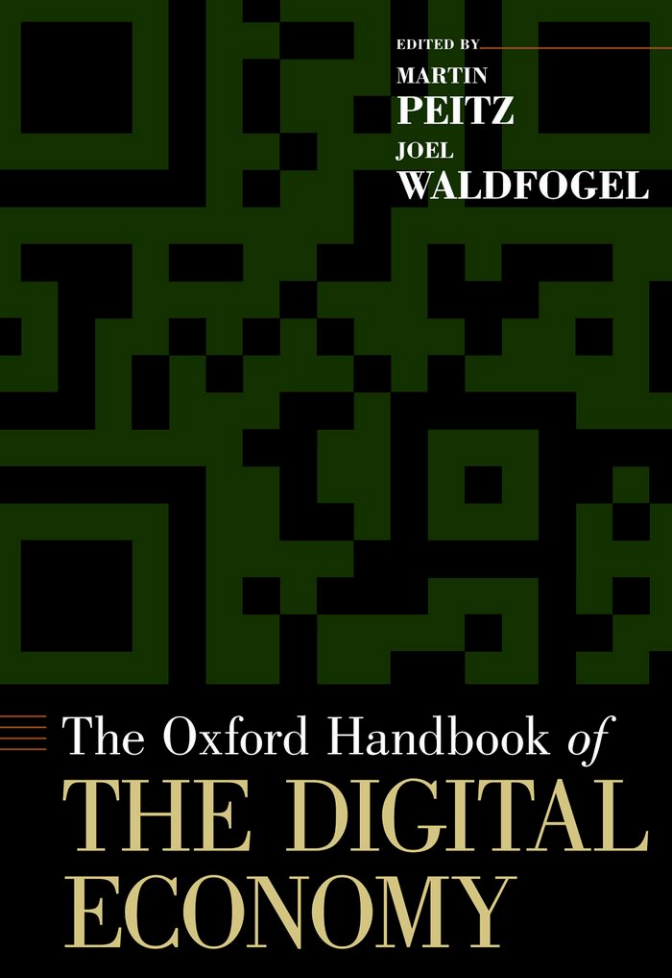




EDITED BY

MARTIN

PEITZ

JOEL

WALDFOGEL

The Oxford Handbook of
**THE DIGITAL
ECONOMY**

THE OXFORD HANDBOOK OF

THE DIGITAL ECONOMY

CONSULTING EDITORS

MICHAEL SZENBERG

LUBIN SCHOOL OF BUSINESS, PACE UNIVERSITY

LALL RAMRATTAN

UNIVERSITY OF CALIFORNIA, BERKELEY EXTENSION

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and

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CONTRIBUTORS

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Alessandro Acquisti is Associate Professor of Information Technology and Public Policy at Heinz College at Carnegie Mellon University.

Ross Anderson is Professor in Security Engineering at the University of Cambridge Computer Laboratory.

Simon P. Anderson is Commonwealth Professor of Economics at the University of Virginia.

Paul Belleflamme is Professor of Economics at Université Catholique de Louvain.

Wilko Bolt is an economist at the Economics and Research Division of De Nederlandsche Bank.

Laura Brandimarte is a PhD candidate in the Public Policy Department (Heinz College) at Carnegie Mellon University.

Sujit “Bob” Chakravorti is the Chief Economist and Director of Quantitative Analysis at The Clearing House.

Luis Cabral is Professor of Economics and Academic Director (New York Center), IESE Business School, University of Navarra and W.R. Berkley Term Professor of Economics at the Stern School of Business at New York University.

Jay Pil Choi is Scientia Professor in the School of Economics at the Australian School of Business, University of New South Wales.

Joseph Farrell is Professor of Economics at the University of California–Berkeley.

Drew Fudenberg is the Frederick E. Abbe Professor of Economics at Harvard University.

Sanjeev Goyal is Professor of Economics at the University of Cambridge.

Shane Greenstein is the Elinor and H. Wendell Hobbs Professor of Management and Strategy at the Kellogg School of Management at Northwestern University.

Ben Greiner is Lecturer at the School of Economics at the University of New South Wales.

Andrei Hagiu is Assistant Professor of Strategy at the Harvard Business School.

Steffen Hoernig is Associate Professor with “Agregação” at the Nova School of Business and Economics in Lisbon.

Lian Jian is Assistant Professor of Communication and Journalism at the Annenberg School, University of Southern California.

Justin Johnson is Associate Professor of Economics at the Johnson Graduate School of Management at Cornell University.

Bruno Jullien is a Member of the Toulouse School of Economics and Research Director at CNRS (National Center for Scientific Research).

Robin S. Lee is Assistant Professor of Economics at the Stern School of Business at New York University.

Ethan Lieber is a PhD student in the Economics Department at the University of Chicago.

Jeffrey K. MacKie-Mason is the Dean of the School of Information at the University of Michigan.

Jose-Luis Moraga-Gonzalez is Professor of Microeconomics at VU University–Amsterdam and Professor of Industrial Organization at the University of Groningen.

Tyler Moore is a postdoctoral fellow at the Center for Research on Computation and Society at Harvard University.

Axel Ockenfels is Professor of Economics at the University of Cologne.

Martin Peitz is Professor of Economics at the University of Mannheim.

Abdolkarim Sadrieh is Professor of Economics and Management at the University of Magdeburg.

Timothy Simcoe is Assistant Professor of Strategy and Innovation at the Boston University School of Management.

Chad Syverson is Professor of Economics at the Booth School of Business, University of Chicago.

Tommaso Valletti is Professor of Economics at the Business School at Imperial College, London.

J. Miguel Villas-Boas is the J. Gary Shansby Professor of Marketing Strategy at the Haas School of Business, University of California–Berkeley.

Joel Waldfogel is the Frederick R. Kappel Chair in Applied Economics at the Carlson School of Management at the University of Minnesota and a Research Associate at the National Bureau of Economic Research.

Matthijs R. Wildenbeest is Assistant Professor of Business Economics and Public Policy at the Kelley School of Business, Indiana University.

INTRODUCTION

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Martin Peitz and
Joel Waldfogel

Digitization—and the Internet—have transformed business and society in the past 15 years. The world’s Internet-connected population has grown from essentially zero in 1995 to 2 billion today.¹ The Internet and digitization have transformed many industries, including retailing, media, and entertainment products. Millions of people spend hours each day accessing and creating information online.

The firms of the digital economy not only affect the daily life of most people in industrialized countries, but they are also highly profitable. Once nearly left for dead, Apple was in 2011 the third most valuable company in the world, with a market valuation of almost \$300 billion US, far above its new-economy forebears Microsoft, IBM, and AT&T at \$239, \$182, and \$174, respectively. Rounding out the newer generation of digital firms are Google, valued at \$192 billion, Amazon at \$77 billion, Facebook at \$83 billion, and eBay at \$36 billion.² While particular valuations fluctuate, it is clear that firms actively in the digital economy are now global players with market capitalizations exceeding those of long-established companies in more traditional industries.

New digitally enabled technologies have facilitated more extensive application of many business strategies that had formerly occupied the attention of economic theorists more than business practitioners. The practices include auctions, price discrimination, and product bundling. Although they have been employed for centuries, they are easier to utilize in digital contexts. Software operating on platforms has heightened the role of platform competition and network effects. The Internet has also provided a stage for many phenomena that would have been difficult for economists—or others—to imagine a few decades ago: large-scale sharing of digital material at sites such as YouTube, Facebook, and Wikipedia. Open-source software reflects related behaviors. Along with many new opportunities, the Internet has also brought some new threats. These include threats to businesses, such as piracy and the security of connected devices, as well as threats to individuals, such as privacy concerns.

These developments have prompted what is, by academic standards, a rapid outpouring of research. This volume is an attempt to describe that work to date,

with the goals of both explicating the state of the literature and pointing the way toward fruitful directions for future research. The book's chapters are presented in four sections corresponding four broad themes: (1) infrastructure, standards, and platforms; (2) the transformation of selling, encompassing both the transformation of traditional selling and new, widespread application of tools such as auctions; (3) user-generated content; and (4) threats in the new digital environment. No chapter is a prerequisite for another, and readers are encouraged to read in any order that aids digestion of the material. A guide to the chapters follows.

The first section deals with infrastructure, standards, and platform competition. In chapter 1, Shane Greenstein provides an overview on Internet infrastructure with a particular emphasis on Internet access and broadband development. There is an interdependence between infrastructure development and changes in industries that rely on this infrastructure. Thus, researchers investigating such industries need a proper understanding of Internet infrastructures. In chapter 2, Joseph Farrell and Timothy Simcoe discuss the economics of standardization, and describe the costs and benefits of alternative ways to achieve compatibility.

The section then turns to issues related to platform competition. The next chapters focus on a number of industries that heavily rely on recent developments in electronic data storage and transmission. In chapter 3, Andrei Hagiu provides a detailed view on developments in business strategies in the market of software platforms. He offers some rich institutional details on this important and rapidly changing industry. In chapter 4, Robin Lee provides a related analysis for the videogame industry. In chapter 5, Wilko Bolt and Sujit "Bob" Chakravorti survey the research on electronic payment systems. In chapter 6, Steffen Hoernig and Tommaso Valletti offer an overview on recent advances on the economics of mobile telephony. All these industries are characterized by mostly indirect network effects. Here, to understand the development of these industries, one has to understand the price and non-price strategies chosen by platform operators. Thus, an important part of the literature relates to the recent theory of two-sided markets. This theme will reappear in a number of other chapters in this handbook. Finally, in chapter 7, Bruno Jullien contributes to our understanding of B2B platforms, drawing on insights from the two-sided market literature. This theory-oriented chapter also helps in formalizing other platform markets.

The second section deals with the transformation of selling. The Internet has transformed selling in a variety of ways: the reduced costs of online retailing threatens offline retailers, widespread availability of information affects competition, digital technology allows the widespread employment of novel pricing strategies (bundling, price discrimination), and auctions are now seeing wide use. In chapter 8, Chad Syverson and Ethan Lieber focus on the interdependence between online and offline retailing. In chapter 9, Jose-Luis Moraga and Matthijs Wildenbeest survey the work on comparison sites. They develop a formal model that reproduces some of the insights of the literature.

Within the general topic of selling, the volume then turns to pricing practices. In chapter 10, Drew Fudenberg and Miguel Villas-Boas start with the observation

that due to advances in information technology, firms have ever more detailed information about their prospective and previous customers. In particular, when firms have information about consumers' past purchase decisions, they may use this information for price discrimination purposes. The focus of the chapter is on the effects of price discrimination that is based on more detailed customer information, both under monopoly and under competition. In chapter 11, Jay Pil Choi surveys the literature on product bundling. This survey focuses on the theory of product bundling and to which extent it is particularly relevant in the context of the digital economy.

The next two chapters address issues related to auctions. In chapter 12, Ben Greiner, Axel Ockenfels, and Abdolkarim Sadrieh provide an overview on Internet auctions. They review theoretical, empirical, and experimental contributions that address, in particular, bidding behavior in such auctions and the auction design in single-unit Internet auctions. In chapter 13, Luis Cabral surveys recent, mostly empirical work on reputation on the Internet, with a particular focus on eBay's reputation system.

The selling section concludes with a contribution on advertising. In chapter 14, Simon Anderson reports on recent developments in online advertising and offers a mostly theory-oriented survey on advertising on the Internet. In particular, he develops a logit model to address some of the most important issues, which draws on the two-sided market literature.

The third section of the book discusses the emergent phenomenon of user generated content on the Internet. In chapter 15, Lian Jian and Jeff MacKie Mason elaborate on user-generated content, an issue that had hardly arisen prior to the digital economy. In chapter 16, Sanjeev Goyal discusses the importance of the theory of social network for our understanding of certain features of the digital economy, in particular, the functioning of social networking sites such as Facebook. In contrast with the literature that postulates network effects in the aggregate, the literature on social networks takes the concrete structure of the network seriously and is concerned in particular with local interactions. In chapter 17, Justin Johnson elaborates on the economics of open source. Partly, open source is user-generated content. An important question concerns the generation of ideas, mostly as software products, with respect to the advantages and disadvantages of open source compared to traditional proprietary solutions. In this context, the openness of a product is also discussed.

The fourth section of the volume discusses threats arising from digitization and the Internet. Chapters 18 and 19 analyze digital piracy. In chapter 18, Paul Belleflamme and Martin Peitz survey the theoretical literature on digital piracy, whereas in chapter 19, Joel Waldfogel surveys the empirical literature. In chapter 20, Alessandro Acquisti and Laura Brandimarte introduce the important issue of privacy in the digital economy. In chapter 21, Ross Anderson and Tyler Moore survey the work on Internet security.

NOTES

1. See Internet World Stats, at <http://www.internetworldstats.com/stats.htm>, last accessed June 10, 2011.
2. See Ari Levy on <http://www.bloomberg.com/news/2011-01-28/facebook-s-82-9-billion-valuation-tops-amazon-com-update1-.html>, last accessed June 2, 2011.

PART I

INFRASTRUCTURE,
STANDARDS, AND
PLATFORMS

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CHAPTER 1

INTERNET INFRASTRUCTURE

SHANE GREENSTEIN

1. INTRODUCTION

Internet infrastructure is a collective term for all the equipment, personnel, and organizations that support the operation of the Internet. The commercial Internet arose after many firms and users voluntarily adopted a set of practices for enabling internetworking, namely, transferring data between local area networks and computer clients. The commercial Internet began to provide many revenue-generating services in the mid-1990s. As of this writing, this network supports a wide array of economic services to more than a billion users, and it continues to grow worldwide.

Generally speaking, four types of rather different uses share the same Internet infrastructure: browsing and e-mail, which tend to employ low bandwidth and which can tolerate delay; video downloading, which can employ high bandwidth and can tolerate some delay; voice-over Internet protocol (IP) and video-talk, which tend to employ high bandwidth and whose quality declines with delay; and peer-to-peer applications, which tend to use high bandwidth for sustained periods and can tolerate delay, but, in some applications (such as Bit-Torrent), can impose delay on others.¹

The precise economic characteristic of Internet infrastructure defies simple description. Like other private assets, sometimes Internet infrastructure is an input in the production of a pecuniary good, and regular investment extends the functionality or delays obsolescence. Like a public good, sometimes more than one user employs Internet infrastructure without displacing another or shaping the

quality of another user's experience. Even when visible, the economic contribution of Internet infrastructure is often measured indirectly at best. Yet in contrast to many private assets or public goods, sometimes many decision makers—instead of one—govern the creation and deployment of Internet infrastructure. Moreover, although market-oriented prices influence the investment decisions of some participants, something other than market prices can shape the extent of investment and the use of such investment.

This array of applications, the mix of economic characteristics, and the economic stakes of the outcomes have attracted considerable attention from economic analysts. This review summarizes some of the key economic insights of the sprawling and vast literature on Internet infrastructure.

The chapter first provides a summary of the emergence of the commercial Internet, providing a review of the common explanations for why its structure and pricing took a specific form. The chapter then describes how the deployment of broadband access and wireless access altered many aspects of Internet infrastructure. It also reviews the importance of platforms, another new development that has changed the conduct of firms providing infrastructure for the commercial Internet. The chapter finishes with a review of the private governance of Internet infrastructure and the role of key government policies.

2. THE EMERGENCE OF THE COMMERCIAL INTERNET

Which came first, the infrastructure or the commercial services? In the popular imagination the Internet emerged overnight in the mid-1990s, with the creation of the commercial browser and services employing that browser. In fact, the Internet developed for more than two decades prior to the emergence of the browser, starting from the first government funding at the Defense Advanced Research Project Agency (DARPA) in the late 1960s and the National Science Foundation (NSF) in the 1980s.² Hence, there is no question that the infrastructure came first. The Internet infrastructure of the mid-1990s employed many components and inventions inherited from the government-funded efforts.

Both the commercial Internet and its government-sponsored predecessor are packet switching networks.³ While the Internet is certainly not the first packet switching network to be deployed, the commercial Internet has obtained a size that makes it the largest ever built.

In a packet switching network a computer at one end packages information into a series of discrete messages. Each message is of a finite size. As part of initial processing by the packet switching system, larger messages were divided into smaller packets. All of the participating computers used the same conventions for messages and packets. Messages successfully traveled between computers when the

sending and receiving computer standardized on the same procedures for breaking up and reassembling packets.

Sending data between two such locations rely on something called “protocols,” standardized software commands that organize the procedures for moving data between routers, computers, and the various physical layers of the network. Protocols also define rules for how data are formatted as they travel over the network. In 1982 one design became the standard for the network organized by DARPA, a protocol known as TCP/IP, which stands for transmission control protocol/internet protocol. Vint Cerf and Robert Kahn wrote the first version, replacing an outmoded protocol that previously had run the DARPA network. Over time the technical community found incremental ways to change the protocol to accommodate large-scale deployment of infrastructure and applications.⁴

TCP/IP had the same role in the government-sponsored and commercial Internet. It defined the “headers” or labels at the beginning and end of a packet.⁵ Those headers informed a computer processor how to reassemble the packets, reproducing what had been sent. By the mid-1980s all Unix-based computing systems built that interface into their operating systems, and in the 1990s virtually every computing system could accommodate it.⁶

An example of TCP/IP in action may help to illustrate its role. Consider sending messages from one computer, say, at Columbia University in New York, to another, say, at Stanford University in California. In the 1980s such an action involved the following steps. A user of a workstation, typically using a computer costing tens of thousands of dollars, would compose the message and then issue a command to send it. This command would cause the data to be broken into TCP/IP packets going to a central mail machine at Columbia and then a destination mail machine at Stanford. In between the central mail machine and destination mail machine the file would be broken down into pieces and completely reassembled at the final destination. During that transmission from one machine to another, the file typically traveled through one or more routers, connecting various networks and telephone lines. At the outbound site it was typically convenient to send through a central mailserver, though that did not need to be true at the inbound site. At the Stanford server, the local computer would translate the data into a message readable on another person’s workstation. If something went wrong—say, a typographical error in the address—then another set of processes would be initiated, which would send another message back to the original author that something had failed along the way.⁷

These technical characteristics define one of the striking economic traits of Internet infrastructure: it requires considerable coordination. First, designers of hardware and software must design equipment to work with other equipment, and upgrades must meet a similar requirement. Second, efficient daily operations of the Internet require a wide array of equipment to seamlessly operate with one another. Indeed, the widespread adoption of TCP/IP partly explains how so many participants coordinate their activity, as the protocol acts as a focal point around which many participants organize their activities at very low transaction costs.

Another key factor, especially during this early period, helped coordinate activity, a principle called *end-to-end*.⁸ This principle emerged in the early 1980s to describe a network where the switches and routers retained general functionality, moving data between computers, but did not perform any processing. This was summarized in the phrase, “The intelligence resided at the edges of the network.” Any application could work with any other at any edge, so long as the routers moved the data between locations.⁹

End-to-end had substantial economic consequences for the deployment of the commercial Internet. At the time of its invention, it was a radical engineering concept. It differed from the principles governing the telephone network, where the switching equipment performed the essential processing activities. In traditional telephony, the end of the network (the telephone handset) contained little functionality.¹⁰ For a number of reasons, this departure from precedent was supported by many outsiders to the established industry, especially researchers in computing. Hence, the Internet encountered a mix of benign indifference and active resistance from many established firms in the communications market, especially in the telephone industry, and many did not invest in the technology, which later led many to be unprepared as suppliers to the commercial Internet in the mid-1990s.¹¹

End-to-end also had large economic consequences for users. First, the Internet largely employed existing capital at many installations with comparatively minor software retrofits, repurposing capital for any new function as long as it remained consistent with end-to-end. Repurposing of existing capital had pragmatic economic advantages, such as lowering adoption costs. It also permitted Internet applications and infrastructure to benefit from the same technical changes shaping other parts of computing.¹² Accommodating heterogeneous installations supported another—and, in the long run, especially important—economic benefit: It permitted users with distinct conditions to participate on the same communications network and use its applications, such as email.

3. THE STRUCTURE OF THE COMMERCIAL INTERNET

As long as the Internet remained a government-managed enterprise, the NSF-sponsored Internet could carry only traffic from the research community, not for commercial purposes.¹³ In the late 1980s the NSF decided to privatize the operations of a piece of the Internet it managed (while the military did not privatize its piece). One key motivation for initiating this action was achievement of economies of scale, namely, the reduction in costs that administrators anticipated would emerge if researchers shared infrastructure with private users, and if competitive processes drove private firms to innovate.¹⁴ The transition reached resolution by the mid-1990s. Many of the events that followed determined the structure of

supply of Internet infrastructure for the next decade and a half, particularly in the United States.¹⁵

The first group of backbone providers in the United States (i.e., MCI, Sprint, UUNET, BBN) had been the largest carriers of data in the NSF network. In 1995 and 1996, any regional Internet service provider (ISP) could exchange traffic with them. At that time the backbone of the US Internet resembled a mesh, with every large firm both interconnecting with every other and exchanging traffic with smaller firms.¹⁶ Some of these firms owned their own fiber (e.g., MCI) and some of them ran their backbones on fiber rented from others (e.g., UUNet).

After 1997 a different structure began to take shape. It was partly a mesh and partly hierarchical, using “tiers” to describe a hierarchy of suppliers.¹⁷ Tier 1 providers were national providers of backbone services and charged a fee to smaller firms to interconnect. The small firms were typically ISPs that ranged in size and scale from wholesale regional firms down to the local ISP handling a small number of dial-in customers. Tier 1 firms did most of what became known as “transit” data services, passing data from one ISP to another ISP, or passing from a content firm to a user. In general, money flowed from customers to ISPs, who treated their interconnection fees with backbone firms as a cost of doing business.

Tier 1 firms adopted a practice known as “peering,” and it appeared to reinforce the hierarchical structure. Peering involved the removal of all monetary transfers at a point where two tier 1 providers exchanged traffic. Peering acknowledged the fruitlessness of exchanging money for bilateral data traffic flows of nearly equal magnitude at a peering point. Hence, it lowered transaction costs for the parties involved. However, because their location and features were endogenous, and large firms that denied peering to smaller firms would demand payment instead, many factors shaped negotiations. As a result, the practice became controversial. An open and still unresolved policy-relevant economic question is whether peering merely reflects a more efficient transaction for large-scale providers or reflects the market power of large suppliers, providing additional competitive advantages, which smaller firms or non-tier 1 firms cannot imitate. The economics of peering is quite challenging to characterize generally, so this question has received a range of analysis.¹⁸

Another new feature of Internet infrastructure also began to emerge at this time, third-party caching services. For example, Akamai, a caching company, would pay ISPs to locate its servers within key points of an ISP’s network. Content providers and other hosting companies would then pay the caching companies to place copies of their content on such services in locations geographically close to users, aspiring to reduce delays for users. Users would be directed to the servers instead of the home site of the content provider and, thus, receive faster response to queries. A related type of service, a content delivery network (CDN), provided similar services for electronic retailers. In general, these became known as “overlays,” and since these were not part of the original design of the noncommercial Internet, numerous questions emerged about how overlays altered the incentives to preserve end-to-end principles.¹⁹

These changes also raised an open question about whether a mesh still determined most economic outcomes. For much Internet service in urban areas the

answer appeared to be yes. ISPs had options from multiple backbone providers and multiple deliverers of transit IP services, and many ISPs multihomed to get faster services from a variety of backbone providers. ISPs also had multiple options among cache and CDN services. Evidence suggests that prices for long-distance data transmission in the United States continued to fall after the backbone privatized, reflecting excess capacity and lower installation costs.²⁰

For Internet service outside of urban areas the answer appeared to be no. ISPs did not have many options for “middle mile” transit services, and users did not have many options for access services. The high costs of supply made it difficult to change these conditions.²¹

Perhaps no factor altered the structure of supply of Internet infrastructure after commercialization more than the World Wide Web: The World Wide Web emerged just as privatization began, and it is linked with a particularly important invention, specifically, the commercial browser. As the commercial Internet grew in the mid to late 1990s, traffic affiliated with the web overtook electronic mail, symptomatic of web applications as the most popular on the Internet. The growth in traffic had not been anticipated at the time NSF made plans for privatizing the backbone, and the subsequent growth in traffic fueled a global investment boom in Internet infrastructure.

Tim Berners-Lee built key parts of the World Wide Web.²² In 1991 he made three inventions available on shareware sites for free downloading: html (hyper-text markup language) and the URL (universal resource locator), a hyper-text language and labeling system that made transfer of textual and nontextual files easier using http (hyper-text transfer protocol). By 1994 and after the plans for commercialization were set and implemented, a new browser design emerged from the University of Illinois. It was called Mosaic, and it was widely adopted in the academic Internet. Mosaic became the model that inspired the founding of Netscape, the first successful commercial browser, which also motivated Microsoft to develop Internet Explorer, which touched off the browser wars.²³

As of this writing most observers expect data traffic to continue to grow at rates in the neighborhood of 40 percent to 50 percent a year.²⁴ This is due to the emergence of another set of applications. Data supporting peer-to-peer and video applications have been growing rapidly after the millennium, each year using larger fractions of available capacity.²⁵

4. PRICING INTERNET ACCESS

The value chain for Internet infrastructure begins with the pricing for Internet access. Households and business establishments pay for access to the Internet.²⁶ ISPs provide that access and use the revenue to pay for inputs, namely, other Internet infrastructure.

In the first half of the 1990s, most commercial ISPs tried the same pricing norms governing bulletin boards. For bulletin boards the pricing structure of the majority of services involved a subscription charge (on a monthly or yearly basis) *and* an hourly fee for usage. For many applications, users could get online for “bursts” of time, which would reduce the size of usage fees.

The emergence of faster and cheaper modems in the mid-1990s and large-scale modem banks with lower per-port costs opened the possibility for a different pricing norm, one that did not minimize the time users spent on the telephone communicating with a server. The emergence of low-cost routines for accessing a massive number of phone lines also contributed to a new norm, because it enabled many ISPs to set up modem banks at a scale only rarely seen during the bulletin-board era.

As the Internet commercialized, two broad viewpoints emerged about pricing. One viewpoint paid close attention to user behavior. Users found it challenging to monitor time online, often due to multiple users within one household. The ISPs with sympathy for these user complaints priced their services as unlimited usage for a fixed monthly price. These plans were commonly referred to as *flat-rate* or *unlimited* plans. Another viewpoint regarded user complaints as transitory. Supporters of this view pointed to cellular telephones and bulletin boards as examples where users grew accustomed to pricing-per-minute. The most vocal supporter for this view was one up-and-coming bulletin-board firm, America Online (now known simply as AOL), which had seemed to grow with such usage pricing.

As it turned out, flat rate emerged as the dominant pricing mode for wireline access. There were many reasons for this. For one, the US telephone universal service policy had long subsidized local landline household calls by charging greater prices for business and long-distance calls than for local calls, which were priced at a flat rate per month in almost every state, with the definition of *local* left to a state regulator. Typically *local* was defined over a radius of ten to fifteen miles, sometimes less in dense urban areas, such as Manhattan. Thus, using local rates reduced household user expenses, thereby making the service more attractive. That enabled an ISP to offer service to nontechnical users, betting the users would *not* stay online for long—in effect, anywhere from twenty to thirty hours of time a month at most. Because such light users did not all dial-in on the same day or at the same time of day, the equipment investment did not need to handle all calls at once. With such light users, an ISP could serve a local area with modem bank capacity at anywhere from one-third to one-quarter the size of the total local service base. A large enough user community could thus share equipment, defraying equipment costs for an ISP offering flat rate. Experiments showed that the economies of scale for defraying equipment costs in this way could support approximately 1000 users in a local point of presence. For many dial-up ISPs, this was not a binding constraint.²⁷

Flat rate succeeded for another reason. Other firms entered into the production of web services, complementing what ISPs provided. Flat rate appealed to users who could then choose among a wide array of media and services.²⁸

Throughout 1996, 1997, and 1998, ISPs experimented with hourly limits and penalties for exceeding these caps, yet most users resisted them. The experiments were motivated by many users who stayed online for far more than twenty to thirty hours a month, thereby ruining the key economics that allowed ISPs to share equipment across many users.²⁹ Most such limits were not particularly binding—involving monthly limits ranging from sixty to one hundred hours. Some ISPs tried offering steep discounts for steep limits, such as \$10 discounts for thirty hours a month. Yet few buyers took them, persisting with the slightly more expensive unlimited contracts, typically priced as \$20 per month. In short, despite all these experiments, flat rate dominated transactions during the dial-up era of the commercial Internet.³⁰

5. THE BROADBAND UPGRADE

As Internet applications became more popular and more widely adopted, users pushed against the bandwidth limits of dial-up Internet access. That motivated some well-placed firms to deploy and offer broadband access as Internet access. The investment of broadband initiated a second wave of investment in Internet infrastructure after the turn of the millennium. That has been coincident with the presence of more capital deepening in business computing and in many facets of the operations to support it.

The two most common forms were cable modem service and (digital subscriber line) (DSL) service. Cable modem service involved a gradual upgrade to cable plants in many locales, depending on the generation of the cable system.³¹ Broadband over telephone lines involved upgrades to telephone switches and lines to make it feasible to deliver DSL. Both of these choices typically supported higher bandwidth to the household than from the household—thus called asymmetric digital subscriber line (ADSL).

Broadband clearly gave users a better experience than dial-up access.³² Broadband provides households with faster Internet service and thus access to better online applications. Broadband also may allow users to avoid an additional phone line for supporting dial-up. In addition, broadband services are also “always on,” and users perceive that as a more convenient service. It is also generally faster in use. A maximum rate of 14.4K (kilobytes per second) and 28.8K were predominant in the mid-1990s for dial-up modems. The typical bandwidth in the late 1990s was 43K to 51K, with a maximum of 56K. DSL and cable achieved much higher maximum bandwidths, typically somewhere in the neighborhood of a maximum rate of 750K to 3M (megabytes per second), depending on the user choices and vendor configuration. Even higher bandwidth became available to some households later.

This story is consistent with Figure 1.1, which summarizes US government surveys of broadband use at households. The first survey questions about broadband

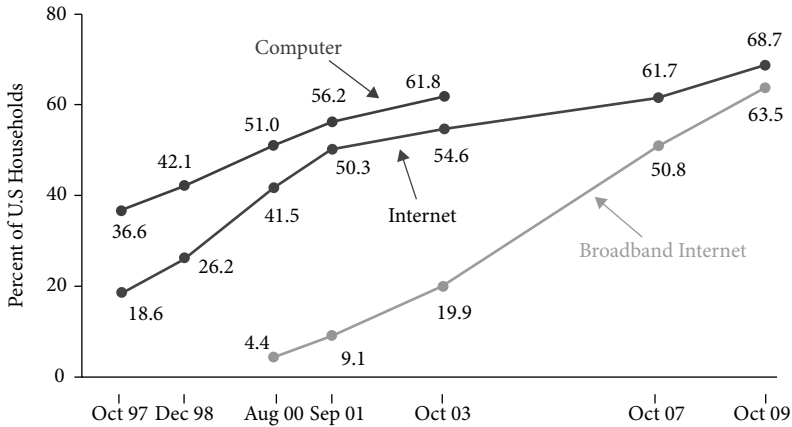


Figure 1.1 Percentage of Households with Computers and Internet Connections, Selected Years, 1997–2009.

Source: NTIA (2010).

use appear in 2000 and show a growth in adoption, reaching close to 20 percent of households in 2003, when these surveys were discontinued for some time.³³ The survey resumed in 2007 and the anticipated trajectory continued, with 50.8 percent of households having broadband in October 2007 and 63.5 percent in October 2009. In the earliest years supply-side issues were the main determinants of Internet availability and, hence, adoption. Cable and telecom operators needed to retrofit existing plants, which constrained availability in many places. In those years, the spread of broadband service was much slower and less evenly distributed than that of dial-up service. Highly populated areas were more profitable due to economies of scale and lower last-mile expenses. As building has removed these constraints, demand-related factors—such as price, bandwidth, and reliability—have played a more significant role in determining the margins between who adopts and who does not.³⁴

Suppliers experimented to find appropriate structures for providing broadband access.³⁵ Countries differed significantly in the extent to which these different delivery channels played a role. Some cable firms built out their facilities to deliver these services in the late 1990s, and many—especially telephone companies—waited until the early to mid-2000s. In some rich countries there was growing use of a third and fourth delivery channel, fiber to the home, and access with mobile modes.³⁶

A similar wave of investment occurred in many developed countries over the first decade of the new millennium. Figure 1.2 shows the subscribers per 100 inhabitants in many countries in 2009.³⁷ Although these numbers must be interpreted with caution, a few facts stand out. A few dozen countries in the Organisation for Economic Co-operation and Development (OECD) experienced substantial adoption of broadband, and many did not. The variance is not surprising. GDP per capita and broadband per capita have a simple correlation of 0.67.³⁸

Figure 1.3 shows the growth of subscribers per 100 inhabitants in the G7, Canada, the United States, United Kingdom, Germany, France, Italy, and Japan,

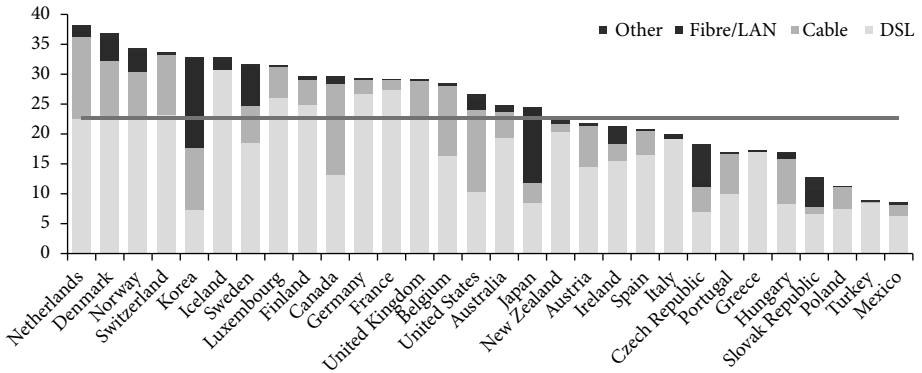


Figure 1.2 OECD Broadband Subscribers per 100 Inhabitants by Technology, June 2009.

Source: OECD.

as well as the entire OECD. Though countries differ in the level of broadband use, the similarities among them are apparent. Adoption of broadband grew in all rich and developed countries.

To give a sense of worldwide patterns, Table 1.1 presents broadband and dial-up adoption for seven countries—the United States, Canada, United Kingdom, Spain, China, Mexico, and Brazil.³⁹ These seven represent typical experiences in the high-income and middle-income countries of the world. The broadband data in Table 1.1 come from Point-Topic, a private consultancy.⁴⁰ One fact is immediately obvious. The scale of adoption in the United States and China far outweighs the scale of adoption in any other country. That occurs for two rather obvious economic reasons. The United States and China have much larger populations than the United Kingdom, Spain, and Canada. Although Mexico and Brazil also have

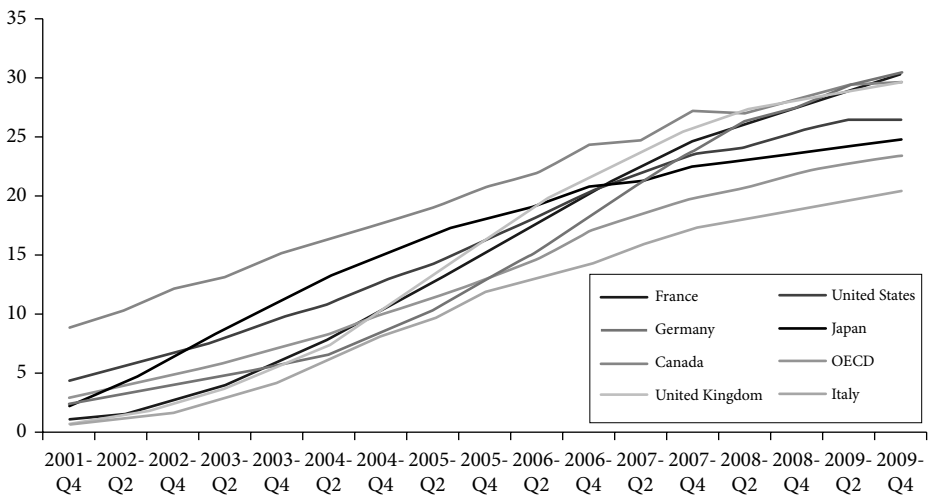


Figure 1.3 Broadband Penetration, G7 Countries.

Source: OECD Broadband Portal, <http://www.oecd.org/sti/ict/broadband>, Table ii.

Table 1.1 Broadband Subscribers from Point Topic

Nation	Broadband Subscribers from Point Topic in Thousands							
	Year							CAGR
	2003	2004	2005	2006	2007	2008	2009	
Brazil	634	1,442	2,671	4,278	5,691	7,509	9,480	47.2%
Canada	3,706	4,829	5,809	6,982	8,001	8,860	9,528	14.4%
China	–	11,385	20,367	30,033	41,778	54,322	68,964	35.0%
Mexico	234	429	1,060	1,945	3,106	4,774	7,836	65.1%
Spain	1,401	2,524	3,444	5,469	7,322	8,296	9,023	30.5%
United Kingdom	960	3,734	7,203	10,983	13,968	16,282	17,641	51.6%
United States	16,042	28,770	37,576	47,489	58,791	67,536	77,334	25.2%

Source: Greenstein and McDevitt (2011)

large populations, those countries had much lower rates of adoption. In short, the general level of economic development is a major determinant of adoption.

The popularity of wireline broadband access has reopened questions about pricing. Following precedent, most broadband providers initially offered unlimited plans. High volume applications—such as video downloading and peer-to-peer applications—placed pressures on capacity, however. This motivated providers to reconsider their flat-rate pricing plans and impose capacity limitations. Hence, a decade and half after the blossoming of the commercial Internet there was considerable speculation about whether video applications would generate a new norm for pricing access. The answer partly depended on the regulatory regime governing Internet access and remains open as of this writing.⁴¹

6. EXPANDING MODES OF ACCESS

In the decade after the millennium another and new set of mobile broadband services began to gain market traction with businesses and households. The first were providers known as wireless ISPs, or WISPs for short. They provided access via a variety of technologies, such as satellite, high-speed WiFi (wireless local area network that uses high-frequency radio signals to transmit and receive data), WiMax (stands for worldwide interoperability for microwave access), and other terrestrial fixed-point wireless delivery modes. These providers primarily served low-density locations where the costs of wireline access were prohibitive.⁴²

Another wireless mode became popular, smart phones. Though smart phones had been available in a variety of models for many years, with Blackberry being among the most popular, it is commonly acknowledged that the category began

to gain in adoption after the introduction of the Apple iPhone in 2007. As of this writing, reports suggest the Apple iPhone, Google Android, and new Blackberry designs dominate this product category for the time being. Yet competitive events remain in flux. Competitive responses organized by Palm, Microsoft, and Nokia have been attempted, and those firms and others will continue to attempt more. In short, irrespective of which firms dominate, smart phones are an important new access point.

The economics of smart-phone use remain in flux as well. It is unclear whether the majority of users treat their smart phones as substitutes to their home broadband use. Smart phones provide additional mobility, and that might be a valuable trait by itself. If smart phones are simply additional services due to mobility, then the economic value of smart phone use should be interpreted one way. If the additional services are partial or complete substitutes, then the economic value of smart phone use should be interpreted another.

Another open question concerns the boundaries between home and business use. With wireline broadband this is less of a question because the destination of the location for Internet access largely identifies its buyer (home or business). Smart phones, however, sell both to home and business, thus obliterating what had been a bright line between these kinds of customers and altering the economic analysis of the adoption decision and its impact. This also alters the potential revenue streams for Internet access, as well as the geographic distribution of use, which could lead to changes in the funding for more Internet infrastructure, as well as the geographic features of investment.

In addition, because of leapfrogging of mobile over fixed broadband in emerging economies, mobile broadband may be the first broadband experience for many people. So not only is it unclear whether mobile broadband substitutes or complements fixed broadband, but the extent of substitutability could vary substantially by country according to each country's stage of infrastructure development.

The changing modes of access opened multiple questions about changing norms for pricing-data services. Wireless data services over cellular networks generally did not use flat-rate pricing and often came with explicit charges for exceeding capacity limits, for example. As of this writing, the norm for these plans was, and continues to be, an open question.

7. STRATEGIC BEHAVIOR FROM SUPPLIERS

When it first deployed, the Internet was called a "network of networks." Although the phrase once had meaning as a description of the underlying configuration of infrastructure, it is misleading today. Leading firms and their business partners view the commercial Internet through the same lens they view the rest of computing. To them, the Internet is a market in which to apply their platform

strategies. In short, the commercial Internet should be called a “network of platforms.”

The list of important platforms in Internet infrastructure is long, and many involve familiar firms, such as Google, Apple, Cisco, Microsoft, and Intel. As a result, the platform strategies of private firms shape the evolution of Internet infrastructure.⁴³ This was one of the biggest changes wrought by introducing private firms into the supply of Internet infrastructure.

Commercial firms regard platforms as reconfigurable clusters and bundles of technical standards for supporting increasing functionality.⁴⁴ From a user perspective, platforms usually look like “standard bundles” of components, namely, a typical arrangement of components for achieving functionality.⁴⁵

Platforms matter because platform leaders compete with others. That usually leads to lower prices, competition in functionality, and accelerated rollout of new services. Many platform leaders also develop new functionality and attach it to an existing platform.⁴⁶ Generally, that brings new functionality to users. A platform leader also might seek to fund businesses outside its area of expertise, if doing so increases demand for the core platform product.⁴⁷ Hence, in the commercial Internet, the strategic priorities of its largest providers tend to shape the innovative characteristics of Internet infrastructure.

Today many observers believe that Google—which did not even exist when the Internet first commercialized in the mid-1990s—has the most effective platform on the Internet. Hence, its behavior has enormous economic consequences. For example, sometimes it makes code accessible to programmers for mash-ups—for example, building services that attract developers and users with no direct way to generate revenue.⁴⁸ Sometimes its investments have direct effects on other Internet infrastructure firms. For example, Google operates an enormous global Internet backbone, as well as CDN and caching services, for its own operations, making it one of the largest data-transit providers in the world today. The firm also uses its platform to shape the actions of others. Many other firms expend considerable resources optimizing their web pages to appear high on Google’s search results, and Google encourages this in various ways. For example, it lets potential advertisers, who will bid in Google’s auction, know which words are the most popular.

Networking equipment provider Cisco is another prominent platform provider for Internet infrastructure, having grown to become a large provider of equipment for business establishments. For many years, Cisco made most of its profit from selling hubs and routers, so the platform strategy was rather straightforward. Cisco aspired to developing closely related businesses, offering users a nearly integrated solution to many networking problems. At the same time, Cisco kept out of service markets and server applications, leaving that to integrators, consultants, and software vendors. That way, Cisco did not compete with its biggest business partners. More recently, however, Cisco branched into consumer markets (with its purchase of Linksys). The firm also has moved into some server (competing with HP) and some software/service areas related to videoconferencing and telepresence

(by purchasing Webex, for example). Cisco no longer draws the boundary where it used to, and it is unclear how wide a scope the firm wants its platform to cover.

Microsoft is perhaps the next best known platform provider whose business shapes Internet infrastructure. In the early 1990s, Microsoft offered TCP/IP compatibility in Windows as a means of enhancing its networking software, as well as to support functionality in some of its applications, such as Exchange. In the mid-1990s, Microsoft offered a browser, partly as a gateway toward developing a broader array of web services, and partly for defensive purposes, to establish its proprietary standards as dominant.⁴⁹ Although Microsoft continues to support these commercial positions and profit from them, the firm has not had as much success in other aspects of its commercial Internet ventures. MSN, search, mobile OS, and related activities have not yielded sustained enviable success (yet). Only the firm's investments in Xbox Live have generated a significant amount of Internet traffic, and it continues to push the boundaries of large-scale multiplayer Internet applications.

Another PC firm, Intel, has an Internet platform strategy. Intel's most important Internet activity came from sponsoring a Wi-Fi standard for laptops under the Centrino brand in 2003.⁵⁰ To be clear, this *did not* involve redesigning the Intel microprocessor, the component for which Intel is best known. It did, however, involve redesigning the motherboard for desktop PCs and notebooks by adding new parts. This redesign came with one obvious benefit: It eliminated the need for an external card for the notebook, usually supplied by a firm other than Intel and installed by users (or OEMs—original equipment manufacturers) in an expansion slot. Intel also hoped that its endorsement would increase demand for wireless capabilities within notebooks using Intel microprocessors by, among other things, reducing their weight and size while offering users simplicity and technical assurances in a standardized function. Intel hoped for additional benefits for users, such as more reliability, fewer set-up difficulties, and less frequent incompatibility in new settings. Intel has helped fund conformance-testing organizations, infrastructure development, and a whole range of efforts in wireless technology. More recently, it has invested heavily in designing and supporting other advanced wireless standards, such as WiMax.

As with many other aspects of commercialization, the importance of platforms is both cause for celebration and a source of concern. It is positive when platform strategies help firms coordinate new services for users. Why is the emergence of platform providers a concern? In short, in this market structure the private incentives of large dominant firms determine the priorities for investment in Internet infrastructure. Under some circumstances dominant platform firms have incentives to deny interconnection to others, to block the expansion of others, and, in an extreme case, to smother the expansion of new functionality by others. When Internet alarmists worry about conduct of proprietary platforms, they most fear deliberate introduction of incompatibilities between platforms, and other conduct to deliberately violate end-to-end principles.⁵¹ Microsoft, AOL, Intel, Comcast, and WorldCom have all shown tendencies toward such behavior in specific episodes.

Open-source advocates who claim that they benefit the Internet usually mean they are preventing defensive activity by leaders with defensive tendencies.

More recently, a range of behavior from Apple, Facebook, Twitter, American Airlines, and the *Wall Street Journal* have raised fears about the “splintering” of the Internet.⁵² Splintering arises when users lose the ability to seamlessly move from one application to another or vendors lose a common set of platforms on which to develop their applications. Steve Jobs’ decision not to support Flash on the iPhone and iPad is one such recent example. So is Twitter’s resistance to having its services searched without payment from the search engines, and Facebook has taken a similar stance. American Airlines refused to let Orbitz see its prices without modifying its website to accommodate new services American Airlines wanted to sell to frequent fliers, and Orbitz refused. As a result, the two companies no longer cooperate. For some time the *Wall Street Journal* has refused to let users search extensively in its archives without a subscription, and its management openly discusses aspirations to gain a fee from search engines, much as Facebook and Twitter did.

What economic incentives lie behind these concerns? In a nutshell the answer begins thusly: no law compels any firm to interconnect with any other. Every firm always has the option to opt out of the Internet, and/or do something slightly less dramatic, such as opt out of commonly used standards, and/or try to get business partners to use proprietary standards. At any given moment, that means a firm in a strong competitive position will have incentives to raise switching costs to its installed base of developers and users, and/or deter migration of users and developers to competing platforms. A variety of strategic options might contribute to such goals, such as designing proprietary standards or blocking others from using a rival firm’s standard designs.

Concerns about splintering also arise because suppliers must cooperate in order to deliver services. Because one party’s cost is another party’s revenue, firms have incentives to sacrifice aspects of cooperation in the attempt to gain rents. Such behavior is not necessarily in users’ interests.

Consider the negotiations between Cogent and Sprint, for example, which broke down after a peering dispute.⁵³ Cogent refused to pay Sprint after Sprint insisted Cogent had not met their obligations under a peering agreement. After a long standoff, Sprint’s management decided to shut down its side of the peering. That action had consequences for users on both networks who did not multi-home, that is, did not use more than one backbone firm. One set of exclusive Sprint users could not reach another set of exclusive Cogent users.⁵⁴ To make a long story short, users of both carriers were angry, and Sprint’s management gave in after a few days. Soon after, the two firms came to a long-term agreement whose details were not disclosed publicly. Notice how inherently awkward the negotiations were: Cogent necessarily interacted or exchanged traffic with the very firm with which it competes, Sprint.⁵⁵

Another set of cases illustrates how the interests of one participant may or may not intersect with the interests of all participants. For example, consider Comcast’s unilateral declaration to throttle peer-to-peer (P2P) applications on its

lines with resets.⁵⁶ This case contains two economic lessons. On the one hand, it partially favors giving discretion to Comcast's management. Management could internalize the externality one user imposes on others—managing traffic for many users' general benefit. That is, P2P applications, like Bit-Torrent, can impose large negative externalities on other users, particularly in cable architectures during peak-load time periods. Such externalities can degrade the quality of service to the majority of users without some sort of limitation or restriction. On the other hand, Comcast's behavior shapes at least one additional provider of applications, future entrepreneurs, many of whom are not present. It would be quite difficult for Comcast and future entrants to reach a bargain because some of them do not even exist yet. Eventually the FCC intervened with Comcast, issuing an order to cease blocking, which led to a court case over its authority to issue such an order. As of this writing, the full ramifications of this court case have not played themselves out.

Another case involving Comcast also illustrates the open-ended nature of the cooperation between firms. In November 2010, Comcast, the largest provider of broadband Internet access in the United States, entered into a peering dispute with Level 3, one of the backbone firms with which it interconnected. Level 3 had made an arrangement with Netflix, a video provider, and this had resulted in Level 3 providing more data to Comcast than Comcast provided to Level 3. Comcast demanded that Level 3 pay for giving more data to Comcast than Comcast gave in return, to which Level 3 agreed.⁵⁷ This agreement was significant because it was the first time an ISP did not pay the backbone provider for transit services, but instead, the provider paid a large ISP to reach end users. As of this writing, it is an open question how common such agreements will become, and whether they will alter the general flow of dollars among Internet infrastructure firms.

8. GOVERNANCE FOR INTERNET INFRASTRUCTURE

A number of organizations play important roles in supporting the design, upgrading, and operations of Internet infrastructure. The most notable feature of the governance of Internet infrastructure are the differences with the governance found in any other communications market.

One notable feature of this structure was the absence of much government directive or mandate. It is incorrect to say that the government was uninvolved: After all, the NSF and Department of Defense both had played a crucial role in starting and sponsoring organizations that managed and improved the operations of the Internet.⁵⁸ Rather, the commercial Internet embodied the accumulation of multiple improvements suggested through a process of consensus in committees, and that consensus depended in large part on private action, what economists call

“private orderings.”⁵⁹ Unlike any other communication network, governments did not play a substantial role in these private orderings.

The organization that governs the upgrades to TCP/IP is the Internet Engineering Task Force (IETF). It was established prior to the Internet’s privatization, and continued as a nonprofit organization after its commercialization. Today it hosts meetings that lead to designs that shape the operations of every piece of equipment using TCP/IP standards.⁶⁰ Many of these decisions ensured that all complying components would interoperate. Today decisions at the IETF have enormous consequences for the proprietary interests of firms.

Standards committees had always played some role in the computer market, and they played a similar role in the shaping of Internet infrastructure. The Institute of Electrical and Electronics Engineers (IEEE), for example, made designs that shaped the LAN market, modem, and wireless data communications markets.⁶¹

Aside from the absence of government mandates, these groups also were notable for the absence of dominant firms. They were not beholden to the managerial auspices of AT&T or IBM, or any other large firm, for example. Though all those firms sent representatives who had a voice in shaping outcomes, these institutions were characterized by divided technical leadership.

That does not imply that all these organizations conducted their business in a similar manner. On the contrary, these forums differed substantially in their conduct.⁶² The World Wide Web Consortium (W3C) offers an illuminating comparison. Berners-Lee forecast the need for an organization to assemble and standardize pieces of codes into a broad system of norms for operating in the hyper-text world. He founded the World Wide Web Consortium for this purpose. In 1994 he established the offices for the W3C in Cambridge, Massachusetts, just in time to support an explosion of web-based services that took advantage of existing Internet infrastructure.

Berners-Lee stated that he had wanted a standardization process that worked more rapidly than the IETF but otherwise shared many of its features, such as full documentation and unrestricted use of protocols. In contrast to the IETF, the W3C would not be a bottom-up organization with independent initiatives, nor would it have unrestricted participation. Berners-Lee would act in a capacity to initiate and coordinate activities. To afford some of these, his consortium would charge companies for participating in efforts and for the right to keep up-to-date on developments.⁶³

The governance structures for the IETF and the W3C also can be compared to what it is not, namely, the next closest alternative for global networking—the Open Systems Interconnection model, a.k.a., OSI seven-layer model. The OSI was a formal standard design for interconnecting networks that arose from an international standards body, reflecting the representation of multiple countries and participants. The processes were quite formal. The network engineering community in the United States preferred their bottom-up approach to the OSI top-down approach and, when given the opportunity, invested according to their preferences.⁶⁴

The lack of government involvement could also be seen in other aspects of the Internet in the United States. For example, the Federal Communications

Commission (FCC) refrained from mandating most Internet equipment design decisions. Just as the FCC had not mandated Ethernet design standards, so it let the spectrum become available for experiments by multiple groups who competed for *wireless* Ethernet standards, which eventually became Wi-Fi. Similarly, the FCC did not mandate a standard for modems other than to impose requirements that limited interference. It also did not mandate an interconnection regulatory regime for Internet carriers in the 1990s.⁶⁵

The US government's most visible involvement in governance has come with its decisions for the Internet Corporation for Assigned Numbers and Names (ICANN), an organization for governing the allocation of domain names. The NSF originally took responsibility for domain names away from the academic community prior to privatizing the Internet, giving it to one private firm. The Department of Commerce established a nonprofit organization to provide oversight of this firm, with the understanding that after a decade ICANN would eventually become a part the United Nations.⁶⁶ This latter transfer never took place, and, as of this writing, ICANN remains a US-based nonprofit corporation under a charter from the Commerce Department.

One other notable innovative feature of Internet infrastructure is its reliance on the behavioral norms and outcomes of open-source projects. This had substantial economic consequences, establishing behavior norms for information transparency that had never before governed the design of equipment for a major communication market. Key aspects of Internet infrastructure embedded designs that emerged from designs that any firm or user could access without restriction, and to which almost any industry participant could make contributions.

One well-known open-source project was Linux, a basis for computer operating systems. It was begun by Linus Torvald in the early 1990s as a derivative, or "fix," to Unix. It was freely distributed, with alternating releases of a "beta" and "final" version. Starting around 1994 to 1995, about the same time as the commercialization of the Internet, Linux began to become quite popular. What had started as a small project caught the attention of many Unix users, who started contributing back to the effort. Many users began converting from proprietary versions of Unix (often sold by the hardware manufacturers) and began basing their operating systems on Linux, which was not proprietary. This movement gained so much momentum that Linux-based systems became the most common server software, especially for Internet servers.⁶⁷

Apache was another early project founded to support and create "fixes" for the HTTP web server originally written by programmers at the National Center for Super Computing Applications (NCSA) at the University of Illinois. By 2006, more than 65 percent of websites in the world were powered by the Apache HTTP web server.⁶⁸ Apache differed from many other open-source organizations in that contributors "earned" the right to access the code. To be a contributor one had to be working on at least one of Apache's projects. By 2006, the organization had an active and large contributor base.

Perhaps the most well-known open source format was the least technical. It originated from something called a *wiki*. Developed in 1995 by Ward Cunningham, a software engineer from Portland, Oregon, wikis can either be used in a closed work group or used by everyone on the Internet. They originally were formed to replicate or make a variation on existing products or services, with the purpose of fixing bugs within the various systems. Accordingly, wikis were first developed and intended for software development but had grown out of that first use and became applied to a multitude of applications. In short, wikis became the essential software infrastructure upon which virtually all major Internet media applications are built.

A particular popular application of wikis, Wikipedia, garnered worldwide attention. In the case of Wikipedia, the format was applied to the development of textual and nontextual content displayed on the Web. It is an online-only encyclopedia. The content is user-created and edited. As its homepage proudly states, it is “The Free Encyclopedia That Anyone Can Edit.” The site has always been free of charge and never accepted advertising.⁶⁹ Wikipedia beat out Microsoft’s Encarta for the honor of the Internet’s top research site in 2005, a position that it has held ever since.⁷⁰

An experimental form of copyright, the creative commons license, spread extraordinarily fast. This license, founded in only 2001, is used by over 30 million websites today.⁷¹ It has begun to play a prominent role in online experimentation and everyday activity. Creative commons licenses help organizations accumulate information in a wide array of new business formats. Flickr is one successful example, having recently passed the milestone of four billion photos on its site. The creative commons license also is employed by numerous Web2.0 initiatives and new media, which support online advertising.

After so much experience with open source it is no surprise that the major participants in Internet infrastructure no longer leave these institutions alone. The standardization organizations find their committees filled with many interested participants, some with explicit commercial motives and some not. These institutions show signs of the stress, chiefly in a slowing down in their decision making, if they reach decisions at all. Perhaps that should also be cause for celebration, since it is an inevitable symptom of commercial success and the large commercial stakes for suppliers.⁷²

9. BROADBAND POLICY

Many governments today, especially outside the United States, are considering making large subsidies for broadband investments. Some governments, such as those of South Korea and Australia, have already done so, making next-generation broadband widely available. Many years of debate in the United States led to the emergence of a National Broadband Plan, released in March 2010.⁷³ Related debates also led to a large European framework for the governance of investments in broadband.⁷⁴

Some of the key issues can be illustrated by events in the United States. At the outset of the commercial Internet, policy favored allowing firms to invest as they please. During the latter part of the 1990s, policy did not restrict the ability of firms to respond to exuberant and impatient demand for new Internet services. After a time, the infrastructure to support those new services became far larger than the demand for services.⁷⁵ After the dot-com boom came to a bust, the United States found itself with excessive capacity in backbone and many other infrastructure facilities to support Internet services.

In the decade after, policy continued to favor privately financed investment. That resulted in a gradual building of broadband. Why did private supply favor gradualism? In short, aside from perceptions of overcapacity, few executives at infrastructure firms would ever have deliberately invested resources in an opportunity that was unlikely to generate revenue until much later, especially ten to twenty years later. Corporate boards would not have approved of it, and neither would stockholders. One of the few firms to attempt such a strategy was Verizon, which unveiled a program to build fiber to the home in the latter part of the first decade after the millennium. Due to low take-up, Verizon did not fully build these services in all its territory.⁷⁶

Most arguments for building next-generation Internet broadband ahead of demand faced large political obstacles. Consider one justification, economic experimentation, namely, better broadband clearly helps experimentation in applications by making users better customers for online ads and electronic retailing. Although the monetary costs are easy to tally, the benefits are not. Relatedly, the costs are focused, but the gains are diffuse, thus making it difficult to show that broadband caused the associated gain, even if, broadly speaking, everyone recognizes that broadband raised firms' productivity and enhanced users' experience. Accordingly, financing broadband would involve a general tax on Yahoo and Amazon and Google and YouTube and other national electronic retailers and application providers who benefit from better broadband. Needless to say, considerable political challenges interfere with the emergence of such schemes. Some countries, such as Korea, have managed to come to such a political agreement, but these examples are the exception, not the rule.

Another set of policies considers subsidizing rural broadband, that is, subsidizing the costs of building wireline broadband supply in high-cost areas.⁷⁷ Since private supply already covers the least costly areas to supply, only a small fraction of potential households benefit from such subsidies, and the costs of subsidizing buildouts are especially high. Such subsidies face numerous challenges. The US National Broadband Plan provides an excellent summary of the issues. Many of the justifications for these subsidies are noneconomic in nature—aimed at increasing civic engagement among rural populations, increasing obtainment of educational goals among children, or increasing the likelihood of obtaining specific health benefits. Accordingly, the decision to provide such subsidies is often a political decision rather than purely an economic one.

Another open question concerns the governance of deployment of access networks. In Europe the governments have chosen a structure that essentially

separates ownership of transmission facilities from ownership of content, and mandates interconnection for many rivals at key points.⁷⁸ In the United States there was a preference for private provision of backbone and access networks in the first decade of the millennium, and a light-handed degree of regulatory intervention, so providers were not required to offer interconnection to others.

No simple statement could characterize the changing norms for regulating Internet infrastructure. For example, as of this writing, at the federal level, there are initiatives under way to adopt formal policies for ensuring the openness of Internet access.⁷⁹ At the local level, there are a range of initiatives by municipalities to provide local access in competition with private suppliers.⁸⁰ There is also effort to limit municipal intervention in the provision of access or deter state limitations on local initiatives.⁸¹

10. SUMMARY

No single administrative agency could possibly have built and managed the commercial network that emerged after the privatization of the Internet. The shape, speed, growth, and use of the commercial Internet after 1995 exceeded the ability of any forecaster inside or outside government circles. The value chain for Internet services underwent many changes after the Internet was privatized. More investment from private firms, and more entry from a range of overlays and new applications, altered nearly every aspect of the structure of the value chain. This evolution occurred without explicit directives from government actors, with only a light hand of directives, and with astonishing speed.

Many key events in Internet infrastructure took place within the United States in the first decade and a half of the commercial Internet, but this appears to be less likely as the commercial Internet grows large and more widespread. While the United States continues to be the source of the largest number of users of Internet services, and the single greatest origin and destination for data traffic, the US position in the global Internet value chain will not—indeed, cannot—remain dominant. That should have enormous consequences for the evolution of the structure of global Internet infrastructure because many countries insist on building their infrastructure according to principles that differ from those that governed the first and second waves of investment in the United States. The boundaries between public and private infrastructure should change as a result, as should the characteristics of the governance and pricing of Internet infrastructure.

It is no surprise, therefore, that many fruitful avenues for further economic research remain open. For example, what frameworks appropriately measure the rate of return in investment in digital infrastructure by public and private organizations? Through what mechanisms does advance Internet infrastructure produce

economic growth, and to which industries in which locations do most of the positive and negative effects flow? What factors shape the effectiveness of different governance structures for open structures, such as those used by the IETF? What is the quantitative value of these novel governance structures?

For the time being there appears to be no cessation in the never-ending nature of investment in Internet infrastructure. Indeed, as of this writing, many questions remain open about the value of different aspects of IT in the long run, and firms continue to explore approaches to creating value. Virtually all participants in these markets expect continual change, as well as its twin, the absence of economic tranquility.

NOTES

1. This is explained in considerable detail in Ou (2008).
2. There are many fine books about these developments, including Abbate (1999), Leiner et al. (2003), and Waldrop (2001).
3. Packet switching had been discussed among communications theorists since the early 1960s, as well as by some commercial firms who tried to implement simple versions in their frontier systems. As has been noted by others, the ideas behind packet switching had many fathers: Paul Baran, J. C. Likelider, Douglas Engelbart, and Len Kleinrock. There are many accounts of this. See, e.g., Quarterman (1989), Abbate (1999), Leiner et al. (2003), or Waldrop (2001).
4. These advances have been documented and analyzed by many writers, including, e.g., Quarterman (1989), Abbate (1999), Leiner et al. (2003), and Waldrop (2001).
5. An extensive explanation of TCP/IP can be found in many publications. Summarized simply, TCP determined a set of procedures for moving data across a network and what to do when problems arose. If there were errors or specific congestion issues, TCP contained procedures for retransmitting the data. While serving the same function of a postal envelope and address, IP also shaped the format of the message inside. It specified the address for the packet, its origin and destination, a few details about how the message format worked, and, in conjunction with routers, the likely path for the packet toward its destination. See, e.g., Leiner et al. (2003).
6. In part, this was due to a DOD requirement that all Unix systems do so, but it also arose, in part, because most Unix users in research environments found this feature valuable.
7. Moreover, a set of fail-safes had been put in place to make sure that one error message did not trigger another. That is, the system avoided the nightmare of one error message generating yet another error message, which generated another and then another, thus flooding the system with never-ending messages.
8. The paper that defined this phrase is commonly cited as Saltzer et al. (1984).
9. As stated by Blumenthal and Clark (2001) in a retrospective look: "When a general-purpose system (for example, a network or an operating system) is built and specific applications are then built using this system (for example, email or the World Wide Web over the Internet), there is a question of how these specific applications and their required supporting services should be designed. The end-to-end arguments suggest that specific application-level functions usually

cannot, and preferably should not, be built into the lower levels of the system—the core of the network.”

10. End-to-end also differed from most large computing systems at the time, such as mainframes, which put the essential operations in the central processing unit. When such computers became situated in a networked environment, there was little else for the terminals to do. They became known as *dumb* terminals. Contrast with end-to-end, which colloquially speaking, located *intelligence* at the edges of the system, namely, in the clients.
11. The economics behind this “surprise” is discussed in some detail in Greenstein (2011).
12. See, e.g., Aizcorbe et al. (2007), Doms and Forman (2005), or the discussion in Forman and Goldfarb (2006).
13. See the discussions in, e.g., Kahn (1995), Kahin and McConnel (1997), and Kesan and Shah (2001), or Greenstein (2010).
14. On many of the challenges during the transition, see Abbate (1999) or Kesan and Shah (2001).
15. For longer discussions about the origins and economic consequences, see, e.g., Mowery and Simcoe (2002a, b), and Greenstein (2011).
16. The term “mesh” is due to Besen et al. (2001).
17. See Friedan (2001).
18. For insights into the incentives to conduct traffic and come to peering agreements, see Besen et al. (2001) and Laffont et al. (2001, 2003).
19. For more on overlays, see Clark et al. (2006).
20. See the evidence in Rosston (2009).
21. This topic has received attention ever since the commercial Internet first began to blossom. See, e.g., Strover (2001) or Downes and Greenstein (2002). A summary can be found in Greenstein and Prince (2007). Also see the discussion in, e.g., the Federal Communications Commission (2010a).
22. See, e.g., Berners-Lee and Fischetti (1999), and Gilles and Cailliau (2000).
23. For more on this story, see Cusamano and Yoffie (2000).
24. Andrew Odlyzko maintains an accessible summary of studies and forecasts of data traffic at <http://www.dtc.umn.edu/mints/>.
25. Anderson and Wolff (2010) present a very accessible version of this argument.
26. Longer explanations for these events can be found in Greenstein (2007a, b).
27. For a longer discussion, see, e.g., Downes and Greenstein (2002, 2007).
28. For a discussion, see, e.g., Goldfarb (2004), Haigh (2007).
29. High usage could happen for a variety of reasons. For example, some technical users simply enjoyed being online for large lengths of time, surfing the growing Internet and Web. Some users began to operate businesses from their homes, remaining online throughout the entire workday. Some users simply forgot to log off, leaving their computers running and tying up the telephone line supporting the connection to the PC. And some users grew more experienced, and found a vast array of activities more attractive over time.
30. One survey of pricing contracts in May 1996 found that nearly 75 percent of the ISPs offering 28K service (the maximum dial-up speed at the time) offered a limited plan in addition to their unlimited plan. That dropped to nearly 50 percent by August. By March of 1997 it was 33 percent, 25 percent by January of 1998, and less than 15 percent by January of 1999. For a summary see Stranger and Greenstein (2007).

31. During the 1990s most cable companies sold access to the line directly to users but made arrangements with other firms, such as Roadrunner or @home, to handle traffic, routing, management, and other facets of the user experience. Some of these arrangements changed after 2001, either due to managerial preferences, as when @home lost its contract, or due to regulatory mandates to give users a choice over another ISP, as occurred after the AOL/Time Warner merger. See Rosston (2009).
32. Download speed may not reach the advertised maxima. In cable networks, for example, congestion issues were possible during peak hours. In DSL networks, the quality of service could decline significantly for users far away from the central switch. The results are difficult to measure with precision.
33. The descriptive results were published in reports authored by staff at the NTIA. See NTIA (2010).
34. In addition to the surveys by Pew, also see, e.g., Savage and Waldman (2004), Rosston et al. (2010), and the summary of Greenstein and Prince (2007). For surveys of business adoption and its variance over geography, see Forman and Goldfarb (2006) and Forman et al. (2005).
35. See, e.g., Crandall (2005).
36. In many areas, fiber to the home was prohibitively expensive for almost all users except businesses, and even then, it was mostly used by businesses in dense urban areas, where the fiber was cheaper to lay. Fiber to the home has recently become cheaper and may become a viable option sometime in the future. See Crandall (2005).
37. OECD Broadband Portal, <http://www.oecd.org/sti/ict/broadband>, Table 1d.
38. OECD Broadband Portal, <http://www.oecd.org/sti/ict/broadband>, Table 1k. For a critique of the US standings in these rankings and insights into how to correct misunderstandings, see Wallsten (2009). Perhaps the biggest issue is the denominator, which is per capita. However, a subscriber tends to subscribe to one line per household. Since US average household size is much larger than average household size in other countries, the figure gives the false impression that US residences have less access to broadband than is actually accurate.
39. These come from Greenstein and McDevitt (2011).
40. For sources, see Greenstein and McDevitt (2009, 2011).
41. See e.g., Federal Communications Commission (2010b), or Marcus (2008).
42. See e.g., <http://www.wispa.org/>. Also see <http://wispassoc.com/>.
43. For interpretations of platform incentives, see e.g., Gawer (2009) or Evans, Haigu and Schmalensee (2006).
44. This is distinct from an engineering notion of a platform. The designers of the Internet deliberately built what they regarded as a computing platform. The inventors employed what they regarded as a balanced and standardized bundle of components to regularly deliver services. This balance reflected a long-standing and familiar principle in computer science. The inventors and DARPA administrators anticipated a benefit from this design: others would build applications, though these inventors did not presume to know what those applications would do specifically.
45. See, e.g., Bresnahan and Greenstein (1999), or Dedrick and West (2001).
46. See, e.g., Bresnahan (1999).
47. See, e.g., Gawer and Cusumano (2002) or Gawer and Henderson (2007).
48. Sometimes Google retains many proprietary features, particularly in its search engine, which also supports a lucrative ad-placement business. Google takes action to prevent anyone from imitating it. For example, the caching, indexing, and underlying engineering tweaking activities remain hidden from public view.

49. See e.g., Cusumano and Yoffie (2000), Bresnahan and Yin (2007), or Bresnahan et al. (2011).
50. For an account of this decision, see Burgelman (2007).
51. For a book with this theme, see, e.g., Zittrain (2009), Lessig (1999), or Von Schewick (2010).
52. Anderson and Wolff (2010) or Zittrain (2009).
53. Once again, this case is explained in detail in Greenstein (2009b).
54. Numerous computer scientists and networking experts have pointed out that both Sprint and Cogent could have adjusted their routing tables in advance to prevent users from being cutoff. Hence, there is a sense in which both parties bear responsibility for imposing costs on their users.
55. It appears that Sprint's capitulation to its user base is, however, evidence that Sprint's management does not have the ability to ignore its users for very long.
56. This case is explained in detail in Greenstein (2009b).
57. For summary, see Adam Rothschild, December 2, 2010, <http://www.voxel.net/blog/2010/12/peering-disputes-comcast-level-3-and-you>.
58. This is especially true of the Internet Architecture Board and IETF, before it moved under the auspices of the Internet Society in 1992, where it remains today. See, e.g., Abbate (1999) or Russell (2006).
59. See Abbate (1999) for a history of the design of these protocols. See Partridge (2008) for a history of the processes that led to the development of email, for example.
60. Simcoe (2007, 2010) provides an overview of the operations at IETF and its changes as it grew.
61. For further discussion see Farrell and Simcoe, chapter 2 of this volume.
62. For example, see, e.g., Chiao et al. (2007).
63. These contrasts are further discussed in Greenstein (2009a).
64. See, e.g., Russell (2006).
65. The latter forbearance was deliberate. On the lack of interference in the design of the Ethernet, see von Burg (2001). On the design of 56K modems, see Augereau, Greenstein, and Rysman (2007). On the lack of regulation for network interconnection, see the full discussions in, e.g., Oxman (1999) or Kende (2000) or the account in Neuchterlein and Weiser (2005). More recent experience has departed from these trends, particularly in principles for regulating last-mile infrastructure. A summary of these departures is in Greenstein (2007b).
66. For a longer explanation of these origins, see, e.g., Kesan and Shah (2001) or Mueller (2002).
67. There is considerable writing about the growth of the production of Linux software and from a variety of perspectives. See, e.g., Dalle, David, Ghosh, and Wolak (2004), Lerner and Tirole (2002), VonHippel (2005), West and Gallagher (2006), or Arora and Farasat (2007). For an account and analysis of how many firms got on the Linux bandwagon, see, e.g., Dedrick and West (2001) or Fosfuri, Giarratana and Luzzi (2005). For further discussion see chapter 17 by Johnson, this volume.
68. For more on the history and operation of Apache, see, e.g., Mockus, Fielding, and Herbsleb (2005).
69. For more information, see Greenstein (2006).
70. For further discussion, see the chapter 3 by Hagiui, this volume.
71. The figures come from the website maintained by the Creative Commons, <http://creativecommons.org/>.

72. For one interesting account of the changing ratio of “suits to beards” at the IETF, see Simcoe (2007, 2010). For an account of the manipulation of hearings at the IEEE, see Mackie-Mason and Netz (2007).
73. Federal Communications Commission (2010a).
74. See Marcus (2008) and Marcus and Elixmann (2008).
75. The overinvestment in Internet infrastructure in the late 1990s had many causes. These are analyzed by, among others, Goldstein (2005), Greenstein (2007b), and Odlyzko (2010).
76. Statistics in the National Broadband Plan, FCC (2010a), seem to indicate that 3 percent of US homes subscribe to this service as of the end of 2009.
77. The economics behind the high cost of providing broadband in low-density locations is explained in detail in Stover (2001), Crandall (2005), and Greenstein and Prince (2007).
78. See Marcus (2008) and Marcus and Elixmann (2008).
79. For a summary, see Goldstein (2005), Nuechterlein and Weiser (2005), Greenstein (2010), and Federal Communications Commission (2010b).
80. See, e.g., Seamans (2010).
81. See, e.g., Baumol et al. (2006).

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CHAPTER 2

FOUR PATHS TO COMPATIBILITY

JOSEPH FARRELL AND
TIMOTHY SIMCOE

1. INTRODUCTION

Compatibility standards are design rules that promote product interoperability, such as the thread size for mechanical nuts and bolts or the communication protocols shared by all Internet devices. Products that adhere to standards should work together well, which produces a range of benefits: users may share information, or “mix and match” components; the cost of market entry declines; and there is a division of labor, thus enabling specialization in component production and innovation. This chapter describes four paths to compatibility—standards wars, negotiations, dictators, and converters—and explores how and when they are used, as alternatives or in combination.

While product interoperability may pose engineering challenges, we focus on issues of economic incentive that arise when its costs and benefits are not evenly distributed. For example, firms that control a technology platform may resist compatibility with other systems or standards that could reduce switching costs for their installed base. Specialized component producers may fight against standards that threaten to “commoditize” their products. Even when support for compatibility is widespread, rival firms may advocate competing designs that confer private benefits because of intellectual property rights, lead-time advantages, or proprietary complements.

Given this mix of common and conflicting interests, we focus on four natural ways to coordinate design decisions (in the sense of achieving compatibility). The first

is decentralized choice, which can yield coordinated outcomes when network effects are strong, even if the resulting process is messy. Negotiations are a second coordination mechanism. In particular, firms often participate in voluntary standard setting organizations (SSOs), which seek a broad consensus on aspects of product design before endorsing a particular technology. A third route to compatibility is to follow the lead of an influential decision maker, such as a large customer or platform leader. Finally, participants may abandon efforts to coordinate on a single standard and instead patch together partial compatibility through converters or multihoming.¹

These four paths to compatibility have different costs and benefits, which can be measured in time and resources, the likelihood of successful coordination for compatibility, and the ex post impact on competition and innovation. Whether these complex welfare trade-offs are well internalized depends on how (and by whom) the path to compatibility is chosen. A full treatment of the compatibility problem would specify the selection process and quantify the relative performance of each path. In practice, although theory clarifies the potential trade-offs, we have limited empirical evidence on the comparative costs and benefits of each path, or the precise nature of the selection process.

Sometimes the choice of a particular path to compatibility is a more-or-less conscious decision. For example, firms can decide whether to join the deliberations of an SSO or follow the lead of a dominant player. A dominant player can decide whether to commit to a standard and expect (perhaps hope) to be followed, or defer to a consensus negotiation. As these examples suggest, it can be a complex question who, if anyone, “chooses” the mechanism, if any, used to coordinate. Some market forces push toward efficiency, but it is not guaranteed. For example, a platform leader has a general incentive to dictate efficient interface standards or to allow an efficient evolution process, but that incentive may coexist with, and perhaps be overwhelmed by, incentives to stifle ex post competition. Likewise, competition among SSOs may or may not lead them toward better policies, and standards wars may or may not tip toward the superior platform.

Sometimes firms will start down one path to compatibility and then veer onto another. For instance, a decentralized standards war may be resolved by resort to an SSO or through the intervention of a dominant firm. Slow negotiations within an SSO can be accelerated by evidence that the market is tipping, and platform sponsors may promote complementary innovation by using an SSO to open parts of their platform. Although theory suggests that certain “hybrid paths” can work well, we know rather little about how different coordination mechanisms complement or interfere with one another.

This chapter begins by explaining something familiar to many readers: how the choice of interoperability standards resembles a coordination game in which players have a mix of common and conflicting incentives. In particular, it explains how compatibility produces broadly shared benefits, and discusses several reasons that firms may receive private benefits from coordinating on a preferred technology. Section 3 describes costs and benefits of our four paths to compatibility. Section 4 examines the selection process and the role of “hybrid” paths to compatibility. Section 5 concludes.

2. COSTS AND BENEFITS OF COMPATIBILITY

When all influential players favor compatibility, creating or upgrading standards involves a coordination problem. When there is but one technology, or when participants share common goals and notions of quality, the solution is primarily a matter of communication that can be solved by holding a meeting, or appointing a focal adopter whom all agree to follow. But if there are several technologies to choose from and participants disagree about their relative merit, it turns a pure coordination game into a battle of the sexes, where players may try to “win” by arguing for, or committing to, their preferred outcome.

Figure 2.1 illustrates the basic dilemma in a symmetric two-player game. As long as $C > B$, the benefits of compatibility outweigh the payoffs from uncoordinated adoption of each player’s preferred technology, and the game has two pure-strategy Nash Equilibria: *both adopt A*, and *both adopt B*. Each player gains some additional private benefit (equal to D) in the equilibrium that selects their preferred technology. When these private benefits are small ($D \approx 0$), many coordination mechanisms would work well. But as D grows large, players will push hard for their preferred equilibrium. Whether these efforts to promote a particular outcome are socially productive depends on a variety of factors about the players’ available actions and the details of the equilibrium selection process. Later, we assume that $D > 0$, and compare the costs and benefits of four broad methods for choosing an equilibrium. But first, this section explains why the payoffs in Figure 2.1 can be a sensible way to model the choice of compatibility standards, particularly in the information and communications technology (ICT) sector.

The benefits of compatibility ($C-B$ in Figure 2.1) come in two flavors: horizontal and vertical. *Horizontal compatibility* is the ability to share complements across multiple platforms, and we call a platform horizontally open if its installed base of complements can be easily accessed from rival systems. Many parts of the Internet are in this sense horizontally open. For example, web pages can be displayed on

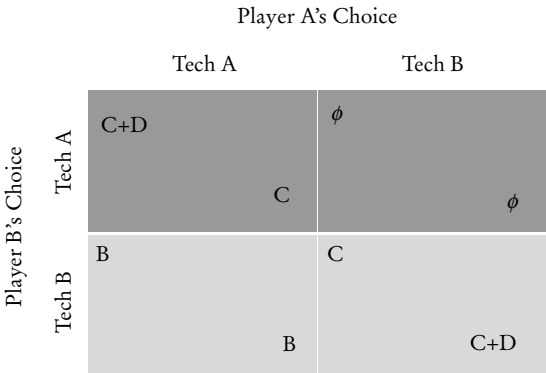


Figure 2.1 Compatibility Choice as a Coordination Game.

competing web browsers, and rival instant messenger programs allow users to chat. Video game consoles and proprietary operating systems, such as Microsoft Windows, by contrast, are horizontally closed: absent further action (such as “porting”), an application written for one is not usable on others.

These distinctions can be nuanced. For example, what if a platform’s set of complements is readily available to users of rival platforms, but at an additional charge, as with many banks’ ATM networks? Similarly, Microsoft may have choices (such as the degree of support offered to cross-platform tools like Java) that affect, but do not fully determine, the speed and extent to which complements for Windows become available on other platforms.

Benefits of horizontal compatibility include the ability to communicate with a larger installed base (direct network effects) and positive feedback between the size of an installed base and the supply of complementary goods (indirect network effects). Katz and Shapiro (1985) analyzed oligopoly with firm-specific demand-side increasing returns. A more recent literature on many-sided platforms (e.g. Rochet and Tirole 2003; Parker and Van Alstyne 2005; Weyl 2010) extends the analysis of indirect network effects by allowing externalities and access prices to vary across different user groups.²

Vertical compatibility is the ability of those other than the platform sponsor to supply complements for the system. We call a platform vertically open if independent firms can supply complements without obtaining a platform leader’s permission.³ For example, the Hush-a-Phone case (238 F.2d 266, 1956), and the FCC’s later *Carterfone* decision (13 F.C.C.2d 420) opened the U.S. telephone network to independently supplied attachments such as faxes, modems, and answering machines. Many computing platforms, including Microsoft Windows, use vertical openness to attract independent software developers. Like horizontal openness, vertical openness can be a matter of degree rather than a sharp distinction. For instance, a platform leader may offer technically liberal access policies but charge access fees.

Vertical compatibility produces several types of benefits. There are benefits from increased variety when vertical compatibility allows users to “mix and match” components (Matutes and Regibeau 1988). Vertical openness also can reduce the cost of entry, strengthening competition in complementary markets. Finally, vertical compatibility leads to a “modular” system architecture and division of innovative labor. Isolating a module that is likely to experience a sustained trajectory of improvements allows other components to take advantage of performance gains while protecting them from the cost of redesign. And when the locus of demand or the value of complementary innovations is highly uncertain, modularity and vertical openness facilitate simultaneous design experiments (Bresnahan and Greenstein 1999; Baldwin and Clark 2000).

The benefits of horizontal or vertical compatibility are often broadly shared but need not be symmetric: there can also be private benefits of having a preferred technology become the industry standard. Such private benefits, labeled D in Figure 2.1, often lead to conflict and coordination difficulties in the search for compatibility.

One important source of conflict is the presence of an installed base. Upgrading an installed base can be costly, and firms typically favor standards that preserve their investments in existing designs. Moreover, platform leaders with a large installed base will favor designs that preserve or increase switching costs, while prospective entrants push to reduce them. For example, in its US antitrust case, Microsoft was convicted of using illegal tactics to prevent Windows users, developers, and OEMs from migrating to the independent standards embodied in the Netscape browser and Java programming language.

Design leads are another source of conflict. Short ICT product life cycles leave firms a limited window of opportunity to capitalize on the demand unleashed by a new standard, and first-mover advantages can be important. Thus, firms may try to block or delay a new standard if rivals have a significant lead at implementation. DeLacy et al. (2006) describe such efforts in the context of Wi-Fi standards development.

In some cases, there is conflict over the location of module boundaries, or what engineers call the “protocol stack.” Since compatibility often promotes entry and competition, firms typically prefer to standardize components that complement their proprietary technology, but leave room for differentiation in areas where they have a technical edge. For example, Henderson (2003) describes how the networking technology start-up Ember allegedly joined several SSOs to prevent new standards from impinging on its core technology.

Conflicts can also emerge when firms own intellectual property rights in a proposed standard, which they hope to license to implementers or use as leverage in future negotiations. Lerner, Tirole, and Strojwas (2003) show that nearly all “modern” patent pools are linked to compatibility standards, and Simcoe (2007) documents a rapid increase in intellectual property disclosures in the formal standard-setting process. While data on licensing are scant, Simcoe, Graham, and Feldman (2009) show that patents disclosed in the formal standards process have an unusually high litigation rate.

Finally, conflicting interests can amplify technological uncertainty. In particular, when technical performance is hard to measure, firms and users will grow more skeptical of statements from self-interested participants about the quality of their favored design.

3. FOUR PATHS TO COMPATIBILITY

Given this mix of conflict, common interest, and incomplete information, choosing compatibility standards can be a messy process. This section considers the performance of four paths to compatibility—standards wars, SSOs, dictators, and converters—in terms of the probability of achieving compatibility, the expected time and resource costs, and the implications for ex post competition and innovation.

We find that economic theory helps articulate some of the complex trade-offs among these paths, but there is little systematic evidence.

3.1. Standards Wars

Standards wars can be sponsored or unsponsored; for brevity we focus here on the sponsored variety, in which proponents of alternative technologies seek to preempt one another in the marketplace, each hoping that decentralized adoption will lead to their own solution becoming a *de facto* standard through positive feedback and increasing returns. Standards wars have broken out over video formats, modem protocols, Internet browsers, and transmission standards for electricity and cellular radio. These wars can be intense when horizontally incompatible platforms compete for a market with strong network effects, which they expect to tip toward a single winner who will likely acquire market power. Much has been written about the tactics and outcomes in such wars, but we do not attempt to discuss them comprehensively here, only to remind the reader of some of the dynamics.⁴

Standards wars often involve a race to acquire early adopters and efforts to manipulate user expectations, as described in Besen and Farrell (1994) or Shapiro and Varian (1998). Preemption is one strategy for building an early lead in an adoption race. Another strategy is to aggressively court early adopters with marketing, promotions, and pricing. Firms may also work to influence users' expectations regarding the likely winner, since these beliefs may be self-fulfilling.⁵

Firms that fall behind in a race for early adopters or expectations may use backward compatibility or bundling to catch up. Backward compatibility jump-starts the supply of complements for a new platform. For instance, many video game platforms can play games written for older consoles sold by the same firm. Bundling promotes the adoption of new standards by linking them to existing technology upgrades. For example, Sony bundled a Blu-ray disc player with the Playstation game console to promote that video format over HD-DVD, and Bresnahan and Yin (2007) argue that Microsoft took advantage of the Windows upgrade cycle to overtake Netscape in the browser wars.

Given the range of tactics used in a standards war, does decentralized technology adoption provide an attractive route to coordination? One social cost is that it will often lead to the emergence of a horizontally closed platform. While one might question the direction of causality (perhaps intractable conflicts over horizontal interoperability lead to standards wars), alternative paths to coordination may produce more *ex post* competition and reduce the risk of stranded investments.

The economic logic of standards wars seems consistent with concerns that markets may "tip" prematurely toward an inferior solution. While many cite the QWERTY keyboard layout (the standard in English-language keyboards) as an example (e.g. David 1990), Liebowitz and Margolis (1990) dispute the empirical evidence and suggest that markets will typically coordinate on the best available technology, as long as the benefits of changing platforms outweigh any switching

costs. It is difficult to find natural experiments that might resolve this debate, for example by randomly assigning an early lead in settings where there are clear differences in platform quality. But even if standards wars typically “get it right” in terms of selecting for quality, coordination problems may affect the timing of standards adoption.⁶

Optimists argue that standards wars are speedy, since participants have strong incentives to race for early adopters, and that fierce ex ante competition offsets any social cost of ex post incompatibility. But competition for early adopters does not always take the form of creating and transferring surplus, and its benefits must be weighed against the costs of stranded investments in a losing platform. Moreover, the uncertainty created by a standards war may cause forward-looking users, who fear choosing the losing platform, to delay commitments until the battle is resolved. For example, Dranove and Gandal (2003) find that preannouncement of the DivX format temporarily slowed the adoption of the digital video disc (DVD). Augereau, Rysman, and Greenstein (2006) suggest that the standards war in 56K modems also delayed consumer adoption.

When it is costly to fight a standards war, participants may seek an escape route, such as some type of truce.⁷ For example, the 56K modem standards war ended in adoption of a compromise protocol incorporating elements of both technologies. The battle between code-division multiple access (CDMA) and time division multiple access (TDMA) cellular phone technology ended in a duopoly stalemate, with each standard capturing a significant share of the global market.

Ironically, these escape routes and stalemates illustrate a final strength of decentralized adoption as a path to compatibility: it can reveal that network effects are weak, or that technologies initially perceived as competing standards can ultimately coexist by serving different applications. For example, Bluetooth (IEEE 802.15) was conceived as a home-networking standard, but ceded that market to Wi-Fi (IEEE 802.11) and is now widely used in short-range low-power devices, such as wireless headsets, keyboards, and remote controls. Similarly, the plethora of digital image formats (JPEG, GIF, TIFF, PNG, BMP, etc.) reflect trade-offs between image quality and compression, as well as compatibility with specific devices. Since “war” is a poor metaphor for the process of matching differentiated technology to niche markets, Updegrove (2007) has proposed the alternative label of “standards swarms” for settings where network effects are weak relative to the demand for variety.

3.2. Standard Setting Organizations

One alternative to standards wars is for interested parties to try and coordinate through negotiation. This process is often called formal or *de jure* standard setting, and typically occurs within consensus standard setting organizations.

There are hundreds of SSOs, and many of these nonprofit institutions develop standards for safety and performance measurement, as well as product

compatibility.⁸ We use a broad definition of SSO that includes globally recognized “big I” standard setters, such as International Telecommunication Union (ITU) and (International Organization for Standardization (ISO); private consortia that manage a particular platform, such as the Internet Engineering Task Force (IETF) and World Wide Web Consortium (W3C); and smaller consortia that focus on a particular technology, such as the USB Forum or the Blu-ray Disc Association.⁹ This definition could even be stretched to include collaborative product-development groups, such as open-source software communities. While the largest SSOs have hundreds of subcommittees and maintain thousands of specifications, small consortia can resemble joint ventures, wherein a select group of firms develop and cross-license a single protocol under a so-called promoter-adopter agreement.

Standards practitioners typically distinguish between consortia and “accredited” standards developing organizations (SDOs). SDOs sometimes receive preferential treatment in trade, government purchasing, and perhaps antitrust in return for adhering to best practices established by a national standards agency, such as the American National Standards Institute (ANSI).¹⁰ Table 2.1 hints at the size and scope of formal standard-setting in the United States by counting entries in the 2006 ANSI catalog of American National Standards and listing the 20 largest ANSI-accredited SDOs.¹¹

SSOs use a consensus process to reach decisions. Though definitions vary, consensus typically implies support from a substantial majority of participants. For example, most accredited SDOs require a super-majority vote and a formal response to any “good faith” objections before approving a new standard. Since SSOs typically lack enforcement power, this screening process may serve as a signal of members’ intentions to adopt a standard, or an effort to sway the market’s beliefs. Rysman and Simcoe (2008) provide some empirical evidence that SSOs’ nonbinding endorsements can promote technology diffusion by studying citation rates for US patents disclosed in the standard-setting process, and showing that an SSO endorsement leads to a measurable increase in forward citations.

Beyond using a loosely defined consensus process and relying on persuasion and network effects to enforce their standards, SSOs’ internal rules and organization vary widely. Some are open to any interested participant, while others charge high fees and limit membership to a select group of firms. Some SSOs have a completely transparent process, whereas others reveal little information. Some SSOs require members to grant a royalty-free license to any intellectual property contained in a standard, whereas others are closely aligned with royalty-bearing patent pools. There has been little empirical research on the internal organization of SSOs, but Lemley (2002) and Chiao, Lerner, and Tirole (2007) examine variation in SSOs’ intellectual property rights policies.

Given SSOs’ heterogeneity, what can we say about the costs and benefits of the consensus process as a path to coordination? Since SSOs encourage explicit comparisons and often have an engineering culture that emphasizes the role of technical quality, there is some reason to expect higher-quality standards than would emerge from a standards war or an uninformed choice among competing

Table 2.1 Major ANSI Accredited SSOs

Acronym	Standards	ICT	Full Name
INCITS	10,503	Y	International Committee for Information Technology Standards
ASTM	8,339	N	American Society for Testing and Materials
IEEE	7,873	Y	Institute of Electrical and Electronics Engineers
UL	7,469	N	Underwriters Laboratories
ASME	7,026	N	American Society of Mechanical Engineers
ANSI/TIA	4,760	Y	Telecommunications Industry Association
ANSI/T1	3,876	Y	ANSI Telecommunications Subcommittee
ANSI/ASHRAE	3,070	N	American Society of Heating, Refrigerating and Air-Conditioning Engineers
AWS	2,517	N	American Welding Society
ANSI/NFPA	2,365	N	National Fire Protection Association
ANSI/EIA	2,011	Y	Electronic Industries Association
ANSI/SCTE	1,803	Y	Society of Cable Telecommunications Engineers
ANSI/AWWA	1,759	N	American Water Works Association
ANSI/AAMI	1,621	Y	American Association of Medical Imaging
ANSI/NSF	1,612	N	National Sanitation Foundation
ANSI/ANS	1,225	N	American Nuclear Society
ANSI/API	1,225	N	American Petroleum Institute
ANSI/X9	940	N	Financial Industry Standards
ANSI/IPC	891	Y	Association Connecting Electronics Industries
ANSI/ISA	872	Y	International Society of Automation
Total ICT	30,786	43%	

Notes: List of largest ANSI accredited Standards Developing Organizations based on a count of documents listed in the 2006 ANSI catalog of American National Standards. The “Standards” column shows the actual document count. The “ICT” column indicates the authors’ judgment as to whether that organization’s primary focus is creating compatibility standards.

technologies. This prediction appears in the stochastic bargaining model of Simcoe (2012), as well as the war-of-attrition model of Farrell and Simcoe (2012), where SSOs provide a quality-screening mechanism.

But technical evaluation and screening for quality can impose lengthy delays, especially when the consensus process gives participants the power to block proposed solutions. A survey by the National Research Council (1990) found that standards practitioners viewed delays as a major problem, and Cargill (2001) suggests that the opportunity costs of delayed standardization explain a broad shift from accredited SDOs toward less formal consortia. Farrell and Saloner (1988) develop a formal model to compare outcomes in a standards war (grab-the-dollar game) to an SSO (war of attrition). Their theory predicts a basic trade-off: the formal consensus

process leads to coordination more often, while the standards war selects a winner more quickly.

SSOs have sought ways to limit deadlocks and lengthy delays. First, some grant a particular party the power to break deadlocks, though such unilateral decisions could be viewed as a distinct route to compatibility (see below). A second approach is to start early in the life of a technology, before firms commit to alternative designs. Illustrating the impact of commitment on delays, Simcoe (2012) shows how delays at the IETF increased as the Internet matured into a commercial platform. But early standardization also has downsides; in particular, private actors have little incentive to contribute technology if they see no commercial opportunity, so anticipatory standards rely heavily on participation from public sector institutions such as academia or government labs.

A third way to resolve deadlocks is to agree on partial or incomplete standards. Such standards often include “vendor-specific options” to facilitate product differentiation. And SSO participants sometimes agree to a “framework” that does not achieve full compatibility but standardizes those parts of an interface where compromise can be reached (thus lowering the ex post cost of achieving compatibility through converters).¹²

Fourth, SSOs may work faster if competing interests are placed in separate forums, such as independent working groups within a large SSO or even independent consortia. Lerner and Tirole (2006) model forum shopping when there is free entry into certification, and show that technology sponsors will choose the friendliest possible SSO, subject to the constraint that certification sways user beliefs enough to induce adoption. This “competing forums” approach works well if there is demand for variety and converters are cheap. But if network effects are strong, forum shopping may produce escalating commitments in advance of a standards war. For example, the Blu-ray and HD-DVD camps each established an independent implementers’ forum to promote their own video format.

Beyond providing a forum for negotiation and certification activities, SSOs are often a locus of collaborative research and development. Thus, one might ask whether selecting this path to coordination has significant implications for innovation. Some forms of innovation within SSOs raise a public goods problem: incentives are weak if all firms have free access to improvements, especially in highly competitive industries. Weiss and Toyofuku (1996) gather evidence of free riding in 10BaseT standards development. Cabral and Salant (2008) study a model where standardization leads to free riding in research and development (R&D), and find that firms may favor incompatibility if it helps them sustain a high rate of innovation. Eisenmann (2008) suggests that SSOs often struggle with “architectural” innovations that span many component technologies, since it is difficult to coordinate the decisions of specialized firms with narrow interests in the outcomes of a particular working group or technical committee.

However, such problems need not prevent all innovation within SSOs. Firms often contribute proprietary technology to open platforms, thus indicating that the benefits of standardizing a preferred technology outweigh the temptation to

free-ride in those cases. Where SSOs encourage horizontal openness, that should encourage innovation in complementary markets by expanding the addressable market or installed base. And while standards can reduce the scope for horizontal differentiation in the market for a focal component, increased competition may stimulate the search for extensions and other “vertical” quality enhancements, as emphasized in Bresnahan’s (2002) analysis of divided technical leadership in the personal computer industry and the quality-ladder model of Acemoglu et al. (2010).

Finally, in horizontally closed platforms, SSOs may encourage innovation by enabling commitments to vertical openness.¹³ In particular, when a platform leader controls some bottleneck resource, small entrants may fear higher access prices or other policy changes that would capture a share of their innovation rents. Platform leaders might solve this hold-up problem by using SSOs to commit to ex post competition (see generally Farrell and Gallini, 1988). For instance, Xerox researchers used an SSO to give away the Ethernet protocol, and Microsoft took the same strategy with ActiveX (Sirbu and Hughes, 1986; Varian and Shapiro, 1998, 254).

One important way SSOs address potential hold-up problems is by requiring firms to disclose essential patents and to license them on reasonable and nondiscriminatory (RAND) terms. These policies seek to prevent patent holders from demanding royalties that reflect coordination problems and the sunk costs of implementation, as opposed to the way that well-informed *ex ante* negotiation would reflect benefits of their technology over the next best solution.¹⁴ Although the “reasonable” royalty requirement can be hard to enforce, and SSOs cannot protect implementers from nonparticipating firms, these intellectual property policies are nevertheless an important method for platform leaders to commit to vertical openness.

In summary, standard setting organizations are a heterogeneous set of institutions linked by their use of the consensus process. This process emphasizes technical performance, and may select for high-quality standards, but can also produce lengthy delays when participants disagree. SSOs also provide a forum for collaborative innovation, and a way for platform leaders to commit to vertical openness in order to promote market entry and complementary innovation.

3.3. Imposing a Standard

A third path to coordination is for someone with sufficient clout to simply impose a standard. This dominant player might be a platform leader, a large customer or complements, or a government agency. A potential advantage of coordination by fiat is speed. In particular, dictators can avoid or resolve deadlocks that emerge in both standards wars and SSOs. Systemwide architectural transitions may also be easier when a *de facto* platform leader internalizes the benefits of a “big push” and is therefore willing to bear much of the cost. However, since dictators are not always benevolent or capable of spotting the best technology, ex post competition, innovation incentives, and technical quality will often depend on who is in charge.

Platform leaders often dictate standards for vertical interoperability. For example, AT&T historically set the rules for connecting to the US telephone network, and IBM has long defined the interfaces used in the market for plug-compatible mainframes. More recently, Apple has maintained tight control over new applications for its iPhone/iPad platform. In principle, platform leaders should have an incentive to use their control over key interfaces so as to organize the supply of complements efficiently. However, fears that incumbent monopolists will block entry or hold up complementary innovators often lead to calls for policy makers to intervene in support of vertically open interfaces.¹⁵

Farrell and Weiser (2003) summarize arguments for and against mandatory vertical openness, and introduce the term internalizing complementary externalities (ICE) to summarize the *laissez faire* position that a platform leader has socially efficient incentives. When ICE holds, a platform leader chooses vertical openness to maximize surplus; open interfaces promote entry and competition in complementary markets, while closed interfaces encourage coordination and systemic innovation. However, Farrell and Weiser note multiple exceptions to the ICE principle, making it difficult to discern the efficiency of a platform leader's vertical policies in practice. For example, a platform sponsor may inefficiently limit access if it faces regulated prices in its primary market; if control over complements is a key tool for price discrimination; if it has a large installed base; or if a supply of independent complements would strengthen a competing platform.¹⁶

In addition to platform leaders, large customers or complementers can act as *de facto* standard setters. For instance, Wal-Mart played an important role in the standardization for radio frequency identification (RFID) chips by committing to a particular specification. Similarly, movie studios played a significant role in resolving the standards war between Blu-ray and HD-DVD. And in some cases, the pivotal "customer" is actually a user group, as when Cable Labs—a consortium of broadcasters—developed the data over cable service interface specification (DOCSIS) protocol for cable modems.

The interests of large complementers and direct customers are often at least loosely aligned with those of end-users, to the extent that their own competitive positions are not threatened. Thus consumers may well benefit from choices made by those powerful players. However, even well-informed powerful players may find it useful to gather information within an SSO before making a decision. Farrell and Simcoe (2012) model this hybrid process, and find that it often outperforms both uninformed immediate random choice and an SSO-based screening process that lacks a dominant third-party.

Government is a third potential dictator of standards. In some cases, the government exerts influence as a large customer. For example, the US Office of Management and Budget Circular A-119 encourages government agencies to use voluntary consensus standards. And in response to requests from the European Union, Microsoft submitted its Open Office XML file formats to ISO. More controversially, governments may use regulatory authority to promote a standard. For example, the US Federal Communication Commission coordinated a switch from

analog (NTSC; named for the National Television System Committee) to digital (ATSC; named for the Advanced Television System Committee) television broadcasting. Sometimes support for standards is even legislated, as in the 2009 stimulus package, which contains incentives for physicians to adopt standardized electronic medical records (but does not take a position on any specific technology).

In general, government involvement can be relatively uncontroversial when there are large gains from coordination and little scope for innovation or uncertainty about the relative merits of different solutions. For instance, it is useful to have standards for daylight-saving time and driving on the right side of the road. Government involvement may also be appropriate when private control of an interface would lead to extreme market power primarily because of severe coordination problems as opposed to differences in quality. But government intervention in highly technical standard-setting processes can pose problems, including lack of expertise, regulatory capture, and lock-in on the government-supported standard.

3.4. Converters and Multihoming

Converters, adapters, translators, and multihoming are ways to reduce the degree or cost of incompatibility. For example, computers use a wide variety of file formats to store audio and video, but most software can read several types of files. Econometricians use translation programs, such as Stat Transfer, to share data with users of different statistical software. Even the Internet's core networking protocols arguably function as a cross-platform converter: as long as all machines and networks run TCP/IP, it is possible to connect many different platforms and applications over a wide variety of physical network configurations.

One benefit of using converters to achieve compatibility is that no single party incurs the full costs of switching. Rather, everyone can choose their preferred system but can also tap into another platform's supply of complements, albeit at a cost and perhaps with some degradation. Since translators need not work in both directions, development costs are typically incurred by the party who benefits, or by a third party who expects to profit by charging those who benefit. And converters can avert the long deadlocks that may occur in a standards war or an open-ended negotiation, since there is no need to agree in advance on a common standard: each platform simply publishes its own interface specifications and lets the other side build a converter (assuming unanimous support for the converter-based solution).

Sometimes users or complementers may join several platforms; such multihoming can resemble a converter solution. For example, most retailers accept several payment card systems, so consumers can pick one and for the most part not risk being unable to transact. Corts and Lederman (2009) show that video game developers increasingly multihome, and argue that multiplatform content explains declining concentration in the console market over time. And instead of seeking a common standard for all computer cables, most machines offer a variety of sockets

to accommodate different connectors such as USB, SCSI, HDMI, and Ethernet. Multihoming preserves platform variety and may align the costs and benefits of horizontal compatibility. However, dedicated converters or coordination on a single platform become more efficient as the costs of platform adoption increase.

Of course, multihoming or converters cannot eliminate conflicting interests, and can open new possibilities for strategic behavior. For example, firms may seek an advantage by providing converters to access a rival's complements while attempting to isolate their own network. Atari tried this strategy by developing a converter to allow its users to play games written for the rival Nintendo platform. However, Nintendo was able to block Atari's efforts by asserting intellectual property based on an encryption chip embedded in each new game (Shapiro and Varian, 1998).

Firms may use one-way converters to create market power on either side of a vertical interface. MacKie-Mason and Netz (2007) suggest that Intel pursued this strategy by including a "host controller" in the USB 2.0 specification, and allowing peripheral devices to speak with the host-controller, but delaying the release of information about the link between the host-controller and Intel's chipsets and motherboards.

Converters can also favor a particular platform by degrading, rather than fully blocking, interoperability. Many computer users will be familiar with the frustrations of document portability, even though most word processors and spreadsheets contain converters that read, and sometimes write, in the file formats used by rival software.

Finally, converters may work poorly for technical reasons. This may be particularly salient for vertical interfaces, since allowing designs to proliferate undercuts the benefits associated with modularity and specialization across components. For example, most operating systems do not provide a fully specified interface for third-party hardware (e.g. printers or keyboards), and the "device driver" software that acts as a translator is widely believed to be the most common cause of system failures (Ganapathi et al., 2006).

In summary, converters are attractive because they preserve flexibility for implementers. However, in a standards war, firms may work to block converters, as Atari did. Firms may also gain competitive advantage by using converters to manipulate a vertical interface. And even when there is little conflict, dedicated compatibility standards may dominate converters for heavily used interfaces, where performance and scalability are important.

4. CHOOSING A PATH

What determines which path to compatibility is followed, or attempted, in a particular case? When will that choice be efficient? While data on the origins of compatibility standards are scant, this section offers some remarks on the selection process.¹⁷

Choosing a path to compatibility can itself be a coordination problem, creating an element of circularity in analysis of this choice. We try to sidestep this logical dilemma by grouping platforms into two categories: those with a dominant platform leader, and shared platforms that default to either collective governance (SSOs) or splintering and standards wars. Eisenmann (2008) suggests that this distinction between shared and proprietary platforms emerges early in the technology life cycle, based on firms' strategic decisions about horizontal openness. In particular, he predicts that platform leaders will predominate in "winner-take-all" markets where network effects are large relative to the demand for variety, multi-homing is costly, and the fixed costs of creating a new platform are substantial.

This life-cycle perspective of platform governance is consistent with the intriguing (though unsystematic) observation that many technologies settle on a particular path to compatibility, even a specific agency, and adhere to it over time. For example, the ITU has managed international interoperability of telecommunications networks since 1865, and JEDEC has been the dominant SSO for creating open standards for semiconductor interoperability (particularly in memory chips) since 1968. Likewise, for products such as operating systems and video game consoles, proprietary platform leadership has been the dominant mode of coordination across several generations of technology.

Nevertheless, there are several well-known cases of dominant firms losing de facto control over a platform. The most famous example is IBM and the personal computer architecture. Other examples include the demise of microcomputing incumbents like Digital Equipment; Google replacing AltaVista as the dominant search engine; Microsoft's well-documented struggles to adapt to the Internet; and the ongoing displacement of the Symbian cellular phone operating system by alternatives from Research in Motion (Blackberry), Apple (iPhone), and Google (Android). Bresnahan (2001) suggests that a key condition for such "epochal" shifts in platform leadership is disruptive technical change at adjacent layers of the larger system, since it is hard to displace a platform leader through direct horizontal competition when network effects are strong. Although this observation certainly accords with the facts of well-known cases, it is not very amenable to formal testing given the infrequent nature of such major shifts.

4.1. Selection and Efficiency

When there is a clear platform leader, the ICE principle suggests that leader will have an incentive to choose an efficient coordination process regarding vertical compatibility. For example, the platform leader might delegate standard-setting activities to an SSO when fears of hold-up impede complementary innovation, but impose a standard when SSO negotiations deadlock.

Unfortunately, the ICE principle is subject to many caveats, bringing back questions about whether a platform leader chooses a particular path for its efficiency or for other reasons such as its impact on ex post competition. For instance, Gawer and

Henderson (2007) use Intel's decision to disseminate USB as an example of ICE, while MacKie-Mason and Netz (2007) argue that Intel manipulated USB 2.0 to gain a competitive advantage. Where one study emphasizes the initial decision to give up control over a technology, the other emphasizes the use of one-way converters to exclude competitors and gain lead-time advantages in complementary markets. These competing USB narratives highlight the difficulty of determining a platform leader's motives.

Without a platform leader, it is less clear why the private costs and benefits of choosing an efficient path to compatibility would be aligned. If firms are *ex ante* symmetric and commit to a path before learning the merits of competing solutions, they would have an *ex ante* incentive to choose the efficient mechanism. But standard setting is typically voluntary, and firms do not commit to abide by consensus decisions. Thus, when asymmetries are present, or emerge over time, firms may deviate to a path that favors their individual interests. While these deviations from collective governance may lead to the "forking" of standards, they do not necessarily block the SSO path, and in some cases the remaining participants can still achieve converter-based compatibility.

Some observers suggest that this chaotic situation can deliver the virtues of both decentralized adoption and collective choice. For example, Greenstein (2010) argues that a proliferation of SSOs combined with widespread independent technical experimentation is a sign of "healthy standards competition" on the commercial Internet. This optimistic view emphasizes the virtues of "standards swarms." When network effects are weak (as the absence of a platform leader might sometimes suggest), and substantial market or technological uncertainty exists, decentralized choice can identify promising standards for a particular niche, with SSOs emerging to facilitate coordination as needed. Unfortunately, there is no guarantee that mixing decentralized adoption with SSOs captures the benefits and avoids the costs of either path in isolation. In particular, either path may lead to a stalemate, and when decentralized adoption is the outside option there is always a danger of stranded investments or selecting the wrong system.

A second optimistic argument holds that new ways to govern shared technology platforms will arise in response to market pressures and technological opportunities. For example, Cargill (2001) and Murphy and Yates (2009) claim that accredited SDOs lost market share to small consortia during the 1980s and 1990s because the SDOs' ponderous decision-making procedures were ill-matched to rapid ICT product lifecycles (see also Besen and Farrell 1991). Smaller and less formal organizations might work faster by relaxing the definition of consensus, taking advantage of new technologies for collaboration, and allowing competing factions to work in isolation from one another. Berners-Lee and Fischetti (1999) cite delays at the IETF as a primary motive for creating the World Wide Web Consortium (W3C), and Figure 2.2 shows that consortia are indeed on the rise.¹⁸

While this evolutionary hypothesis is intriguing, it is not obvious that organizational experimentation and competition will evolve an efficient path to compatibility. Simcoe (2012) shows that consortia still experience coordination delays when participants have conflicting interests over commercially significant technology.